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RESEARCH ARTICLE

TrustTrace: Provenance-Aware Tracing of Indirect Prompt Injection Propagation in LLM-Assisted Learning Workflows

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ABSTRACT Indirect prompt injection can persist through retrieval, generation, and revision after the visible payload disappears. Existing defenses primarily detect suspicious inputs or constrain immediate actions, but they provide less evidence about which source influenced which final-artifact span through a temporally valid route. This paper presents TrustTrace, a provenance-aware framework for source-to-artifact tracing in LLM-assisted learning workflows. TrustTrace constructs a heterogeneous temporal graph over sources, retrieved contexts, prompts, responses, revision operations, and artifact spans. A dual-flow encoder separates legitimate evidence transmission from unauthorized control transmission, while differentiable reachability supports source attribution, contaminated-span localization, activation-stage estimation, and complete-path recovery. A bounded counterfactual verifier performs fact-preserving neutralization, suffix replay, calibrated set-valued attribution, and abstention. The primary pipeline uses procedurally separated GPT-5.5 profiles; construct validity is therefore tested with independent human ratings, a Qwen3-235B-A22B-Instruct generation set, and an independent Qwen3 neutralization-and-judging stack. On the independently audited EduTrace-IPI subset, TrustTrace improves strict Complete-Path F1@3 from 0.631 to 0.684 over the architecture-matched terminal-supervision baseline and from 0.659 to 0.716 after disagreement-flagged workflows are excluded. Human ratings of 800 stratified artifact spans correlate with the continuous contamination score at Spearman $\rho = 0.78$, and an independent Qwen3 verifier preserves most of the primary result (Span-F1 0.731 and Complete-Path F1@3 0.671). On Qwen3-generated workflows, TrustTrace obtains Span-F1 0.681 and Complete-Path F1@3 0.598 without retraining. Binary detection gains remain comparatively small, and specialized guards remain stronger on selected native input- or action-level metrics. The evidence supports TrustTrace as a complementary forensic layer for calibrated span and path tracing rather than a replacement for preventive defenses.

INDEX TERMS Indirect prompt injection, provenance graph, large language model security, attack propagation tracing, counterfactual verification.

I. INTRODUCTION

Large language models (LLMs) are increasingly used as retrieval assistants, conversational tutors, writing collaborators, feedback generators, and document-analysis agents in learning systems [1]. A realistic workflow may retrieve external documents, synthesize evidence, generate an initial response, apply automated or human revision operations, and assemble an essay, report, solution, or study note. The primary datasets in this study use fully logged system-generated stages, while a separate HumanEdit-Audit set contains controlled free-form revisions by blinded editors. This extended process enlarges the attack surface because webpages, retrieved passages, shared knowledge bases, and tool outputs may contain instructions that the LLM treats as executable commands rather than untrusted data.

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Indirect prompt injection (IPI) exploits this instruction–data ambiguity by embedding unauthorized directives in content that the system is expected to retrieve or process. Such directives may redirect a task, suppress evidence, manipulate a conclusion or citation, disclose information, or trigger an unintended action [2], [3]. Existing benchmarks demonstrate the practical impact of IPI in document-processing, retrieval-augmented, and tool-using settings [4], [5], [6]. Most evaluations, however, remain event-centered: an attack is declared successful when the model follows an injected instruction, invokes an unsafe tool, or emits an attacker-specified target in an immediate response.

Event-level success does not resolve persistent semantic influence. A revision can remove the visible payload while preserving a biased claim, omitted counterargument, fabricated citation, or distorted evidence selection. Detecting an instruction in a source therefore does not establish adoption, and detecting a suspicious final sentence does not identify its origin or intermediate route. Semantic similarity is also insufficient because legitimate evidence use creates strong source–output dependence. The relevant audit question is narrower and more demanding: which source affected which artifact span, through which temporally valid transformations, and does the dependency remain when the suspected control component is removed while useful facts are preserved?

Input detectors, structured prompting, security alignment, and runtime guards address important preventive objectives [7], [8], [9], [10], [11], [12], [13]. These methods are not generally designed to reconstruct a source-to-span path after multiple rounds of text production. Provenance-aware systems such as ARGUS and counterfactual systems such as AttriGuard provide the closest conceptual foundations [14], [15], but their protected outputs are agent decisions and tool invocations rather than compositional textual artifacts. A learning artifact can inherit both valid evidence and unauthorized control from the same source, so a usable tracer must separate these two forms of transmission rather than treating every dependency as contamination.

We propose TrustTrace, a provenance-aware framework for tracing source-to-artifact IPI propagation in LLM-assisted learning workflows. TrustTrace represents each workflow as a heterogeneous temporal graph over source documents, retrieved passages, trusted prompts, response spans, revision operations, and final-artifact spans. It learns a content-flow channel for legitimate evidence inheritance and a control-flow channel for task redirection, suppression, citation manipulation, and related unauthorized influence. Differentiable path reachability supplies a shared structural basis for source attribution, contaminated-span localization, activation-stage estimation, and explicit path recovery. Workflow-level propagation detection is retained as an auxiliary screening output rather than the central claim.

A bounded verifier then neutralizes the predicted control component, preserves task-relevant propositions, and replays only the affected workflow suffix. The verifier reports a replay-verified dependency, not a causal effect on learner cognition. It returns calibrated candidate-source sets and abstains when interventions disagree, several sources remain indistinguishable, or the trace crosses an unreplayable boundary. The primary pipeline uses procedurally separated profiles of the same GPT-5.5 base service, so prompt separation alone is not treated as model-family independence. We therefore add three external anchors: human ordinal contamination ratings, an independent Qwen3-235B-A22B-Instruct neutralization-and-judging stack, and a test-only Qwen3-generated workflow set. These experiments quantify rather than merely disclose the remaining shared-model dependence.

The educational setting is technically consequential because instructional documents legitimately contain commands, questions, rubrics, worked examples, and procedural language. This creates an over-defense risk for generic instruction detectors. The current evaluation remains trace-level: HumanEdit-Audit tests free-form editor revisions, but none of the datasets is a longitudinal classroom log. The paper therefore makes no claim about learner intent, cognition, misconduct, or educational outcomes.

Table 1 maps each failure mechanism to a design choice and a falsification test. The central evidential comparison is architecture matched: TrustTrace is compared with the same temporal graph trained without edge and path supervision. Native security baselines are retained to establish complementarity, with adapter-derived outputs explicitly distinguished from each method's original protection target.

The main contributions are as follows.

- We formulate IPI auditing as source-to-artifact propagation tracing, separating exposure, activation, retention, and final-artifact contamination. The primary outputs are malicious-source attribution, contaminated-span localization, activation-stage estimation, and complete intermediate-path recovery; binary detection is retained only as an auxiliary screening output.
- We develop a dual-flow temporal provenance graph that separates legitimate evidence inheritance from unauthorized control transmission. Recursive temporal reachability and retention modeling require endpoint predictions to be supported by coherent intermediate structure, while the stated noisy-OR approximation is evaluated under long and convergent paths.
- We introduce bounded replay verification with fact-preserving neutralization, source-set calibration, and abstention. Verification failures are separated into fact loss, residual control, accepted-neutralizer disagreement, unrelated replay drift, coalition ambiguity, and unreplayable human-edit boundaries.
- We provide leakage-controlled evaluation with independently adjudicated source, span, edge, and path labels; disagreement-stratified path analysis; human validation of the continuous contamination score; faithful text-artifact adaptations of ARGUS and AttriGuard;

TABLE 1. Failure-mechanism, design, and evidence mapping. The last column identifies the experiment intended to falsify each design claim.

Failure mechanism	Why a conventional method can fail	TrustTrace response	Falsification evidence
Exposure is not adoption	A source detector flags suspicious text even when the model ignores it	Joint source-to-span path and contamination decision	Source-only/artifact-only controls and clean instructional false-positive rate
Evidence/control conflation	Relevant sources and injected sources can both be semantically similar to the artifact	Separate content-flow and control-flow states	Single-flow ablation, benign-instruction hard negatives, and topic/LLM-style probe control (Table 18)
Revision erases wording	Copying, paraphrasing, and summarization can preserve an objective after lexical cues disappear	Temporal encoding and edge-specific retention gate	Edit-conditioned retention, shuffled-stage control, and path-length analysis
Correct endpoints, wrong route	Source and span predictions can be correct while the intermediate provenance path is wrong	Differentiable temporal reachability with edge and path supervision	Edge-AUPRC, complete-path F1@3, normalized path edit distance, and path coverage
Source ambiguity	Several sources may support or jointly induce the same final claim	Counterfactual replay, set prediction, and abstention	Neutralizer audit, multi-source coalition tests, source-set coverage, and selective error
Shared-model evaluator bias	Generation, neutralization, and judging can inherit one model family's blind spots	Independent human scoring and a Qwen3 generation/judge/verifier replication	Cross-family score agreement, cross-family tracing, and human ordinal-score calibration
Future-information leakage	Completed artifacts can make earlier dependency construction unrealistically easy	Strict directed acyclic graph construction using only past information	Intentionally invalid future-edge positive control

an independent Qwen3 judge/verifier; Qwen3-generated and human-edited test sets; adaptive attacks; and record-level reproducibility artifacts.

II. RELATED WORK

A. INDIRECT PROMPT INJECTION ATTACKS AND BENCHMARKS

Prompt injection arises because LLM-integrated applications process instructions and data within a shared natural-language context. Early work showed that this design permits external content to override application instructions, thereby enabling remote manipulation of LLM-integrated systems [2]. Subsequent work formalized the attack space and provided systematic comparisons of attack and defense strategies across models and tasks [3]. These studies shifted the discussion from isolated demonstrations toward repeatable security evaluation. More recent attacks increase stealth by introducing the malicious objective through a gradual topic transition, while memory- and knowledge-base poisoning can cause compromised information to be retrieved only under optimized triggers [16], [17]. Agent Security Bench further broadens evaluation across heterogeneous agent scenarios, tools, attack classes, and defenses [18].

Indirect prompt injection specifically places the malicious instruction in an external source rather than in the user's direct input. BIPIA evaluates this setting across several content formats and tasks and shows that models have difficulty distinguishing informational content from actionable instructions [4]. InjecAgent extends the threat to tool-integrated agents and covers attack goals such as direct user harm and private-data exfiltration [5]. AgentDojo provides a dynamic environment that evaluates both benign task utility and security under attacks embedded in tool outputs [6]. Together, these benchmarks established the importance of measuring security and utility jointly.

Most existing benchmarks label success at the response or action level: an attack succeeds when the model follows the malicious instruction, emits a target response, leaks specified information, or invokes an attacker-desired tool [3], [5]. AgentDojo similarly evaluates prompt-injection attacks through task completion and attacker-defined outcomes in tool-mediated agent execution [6]. This event-based design is appropriate for evaluating immediate compromise, but it does not represent delayed semantic propagation through a text-production workflow. A contaminated response can be revised several times before becoming part of a final artifact, and the final text may preserve the attack objective without retaining the injection wording. TrustTrace complements existing benchmarks by treating the propagation state at each workflow stage and the source-to-span path as first-class labels.

Another emerging concern is benchmark saturation and adaptive validity. AgentDojo was designed as an extensible environment rather than a fixed test set, reflecting the fact that attack and defense strategies co-evolve [6]. Adaptive attacks have subsequently bypassed multiple defenses that appeared effective under static evaluation [19]. This evidence suggests that a propagation detector should not be evaluated only on known imperative patterns. Our setting therefore includes unseen payloads, paraphrased control objectives, benign instructions embedded in legitimate educational material, and attacks optimized to preserve semantic influence while avoiding surface-level detection.

B. DETECTION, ALIGNMENT, AND RUNTIME DEFENSE

Existing defenses can be broadly organized into input-level detection and sanitization, model-level instruction alignment, and system-level runtime enforcement. Detection-based approaches attempt to recognize injected instructions before or while the LLM processes external content. Chen et al. study whether IPI can be detected and removed and evaluate segmentation- and extraction-based removal strategies [7]. InstructDetector instead uses intermediate model states and gradients as features for instruction detection [8]. Such approaches can be deployed as filters and may be effective when malicious content has recognizable behavioral or

representational signatures. DataSentinel strengthens this direction by formulating detection as a minimax game in which the detector is trained against strategically adapted injections [20].

However, input detection alone cannot determine whether suspicious text actually influenced a later artifact. Recent prompt-guard evaluations show that benign inputs containing instruction-like trigger words can be falsely classified as attacks, producing substantial over-defense [21]. Instruction-detection methods can identify behavioral-state changes caused by external directives, but their primary decision remains whether the current input contains an actionable instruction [8]. Educational and instructional documents frequently contain imperative language, quoted commands, exercises, rubrics, and procedural steps. Flagging all instruction-like text therefore creates over-defense, while an attacker can paraphrase a control objective into less conspicuous language. The distinction required in this work is not simply instruction versus non-instruction, but legitimate content dependence versus unauthorized control dependence. TrustTrace consequently uses source detection only as one signal within a propagation graph and evaluates the downstream survival of influence.

Model-level defenses aim to make the underlying LLM respect a trust hierarchy. StruQ separates trusted prompts and untrusted data into structured channels and trains the model to follow instructions only from the prompt channel [9]. SecAlign uses preference optimization to favor outputs that follow the legitimate instruction over outputs that comply with an injected instruction [10]. These methods directly target the instruction–data confusion that enables prompt injection and can improve robustness without introducing a separate detector at every application boundary. Complementary model- and prompt-level strategies mark untrusted content to preserve source boundaries and explicitly train models to prioritize privileged instructions [22], [23].

Their protection nevertheless depends on the deployed model and its alignment behavior. Closed-source systems may not permit retraining, and adaptive attacks can target the learned decision boundary or the surrounding application logic. In addition, model-level refusal does not provide a post hoc explanation when compromised content has already entered a multi-turn workflow. TrustTrace is therefore positioned as an application-level detection and forensic layer that can operate over recorded workflow traces and can complement, rather than replace, aligned models.

System-level defenses move enforcement outside the generative model. Task Shield checks whether proposed instructions and actions contribute to the user's task [11]. CaMeL extracts control and data flow from the trusted query and uses capabilities to prevent untrusted data from changing protected program flow [12]. RTBAS adapts information-flow control to tool-based agents and uses dependency screening to limit unnecessary taint propagation [13]. These approaches are important because they reduce reliance on the same model that is being protected.

The principal object of these systems is usually an executable action, tool invocation, or information release. In contrast, the output of an LLM-assisted learning workflow is a text artifact whose risk can be distributed across claims and paragraphs. Preventing all low-integrity content from influencing the artifact would also remove legitimate retrieved evidence. The present work therefore focuses on selective tracing: it models which information was used as evidence and which information exerted unauthorized control, then localizes the residual influence after generation and revision.

C. PROVENANCE, INFORMATION FLOW, AND COUNTERFACTUAL ATTRIBUTION

Provenance records where data originate and how they are transformed. In grounded text generation, attribution methods connect generated statements to supporting documents or source spans. LAQuer, for example, localizes source spans that support user-selected output spans, enabling more focused verification than sentence-level citation alone [24]. This line of work provides useful mechanisms for source–output alignment and motivates fine-grained span attribution. Related grounded-generation resources evaluate citation support and word-level hallucination in retrieved-context outputs [25], [26]. RAGAs and ARES additionally assess context relevance, answer faithfulness, and answer relevance across retrieval-augmented generation pipelines [27], [28]. These methods make evidence dependence measurable, but they do not distinguish legitimate support from adversarial control.

Security provenance requires an additional distinction. Structured-query defenses demonstrate the value of separating trusted prompts from untrusted data channels [9], while system-level designs such as CaMeL explicitly extract control and data flows to prevent untrusted observations from changing program execution [12]. A malicious source may nevertheless contain valid factual content and an unauthorized instruction at the same time. The facts may legitimately support the answer, whereas the instruction may alter the answer's goal, framing, evidence selection, or citation behavior. A single undifferentiated provenance edge cannot represent these two effects. TrustTrace therefore introduces separate content-flow and control-flow relations so that evidence inheritance does not automatically imply attack propagation and semantically transformed control can still be detected when lexical overlap is weak.

Information-flow defenses for LLM agents similarly track whether low-integrity or confidential data influence protected actions. CaMeL represents expected control and data flow explicitly, while RTBAS selectively propagates labels according to inferred dependencies [12], [13]. More recent provenance-aware agent defenses construct influence

graphs that connect untrusted context to agent decisions. ARGUS audits whether a decision is justified by trustworthy evidence in context-dependent agent tasks [14]. These methods demonstrate the value of graph-based provenance for runtime security.

TrustTrace differs from action-centered defenses in both the protected object and the temporal scope. Task Shield verifies whether instructions and tool calls remain aligned with the user's task [11], and IPIGuard constrains execution through a planned tool-dependency graph [29]. AttriGuard further applies counterfactual causal attribution to determine whether an untrusted observation drives a particular tool invocation [15]. TrustTrace instead traces influence through retrieved passages, generated spans, revision operations, and final-artifact spans. The graph must therefore represent transformation operators such as copying, summarization, paraphrasing, deletion, and logged revision operations, while tolerating cases in which one source contributes both benign evidence and adversarial control. This source-to-artifact setting requires span-level outputs and stage-wise propagation labels that are not provided by action-centered provenance graphs.

Counterfactual replay provides a way to test whether an inferred dependency is more than correlation. AttriGuard evaluates whether untrusted observations drive a tool invocation by re-executing the agent after attenuating potential control channels [15]. This is closely related to the verification objective of TrustTrace and establishes that intervention-based attribution can improve IPI defense. The present work extends the idea to compositional text propagation. Instead of asking whether one tool call survives, it selectively removes the control component of a suspected source, preserves task-relevant facts, and replays the downstream workflow suffix to determine which intermediate and final spans change. This design also addresses a confound in naive source deletion: removing the whole source may change the output because useful evidence is missing, not because an attack has been neutralized.

Because LLM outputs are stochastic and semantically non-unique, counterfactual differences cannot be interpreted using exact string equality. TrustTrace therefore combines semantic target tests with repeated replay and calibrated attribution. It reports a candidate source set or abstains when several sources are indistinguishable. This emphasis on uncertainty is important for forensic use, where a confidently wrong source attribution may be more harmful than an unresolved case.

D. LLM-ASSISTED LEARNING INTEGRITY AND POSITION OF THIS WORK

Educational LLM agents increasingly use memory, retrieval, tools, planning, and multi-agent interaction to support feedback, curriculum design, knowledge tracing, error correction, and personalized learning [1]. These capabilities make the learning setting technically similar to other agentic systems, but the protected outcome is different. A learning artifact is often developed through multiple rounds of model generation and learner revision. Its integrity depends not only on whether the system avoided an immediately harmful action, but also on whether claims, evidence, citations, and conclusions remain aligned with the learner's task and the trusted source set.

Research on educational LLMs has emphasized hallucination, overreliance, privacy, bias, and integration with existing learning environments [1]. Prompt injection adds an external integrity threat: the model may produce a coherent and apparently well-supported artifact whose content was redirected by an attacker-controlled source. This threat should not be reduced to student misconduct or psychological reliance. It can be studied as a system-security problem with controlled sources, recorded workflow traces, and automatically generated propagation labels, thereby avoiding subjective judgments about a learner's intent.

The proposed work occupies the intersection of IPI defense, provenance learning, and content-integrity analysis. It differs from input detectors by evaluating downstream influence; from aligned models by providing application-level forensic tracing; from tool-centered runtime defenses by protecting textual artifacts; and from standard source attribution by separating evidence flow from adversarial control flow. The learning system provides a concrete and consequential workflow, while the underlying source-to-artifact formulation is applicable to other human–LLM production settings such as report writing, policy analysis, literature synthesis, and collaborative document preparation.

III. METHODOLOGY

A. WORKED EXAMPLE AND AUDIT OUTPUTS

Consider a task that asks for a balanced comparison of two policy positions. Source d_2 contains relevant statistics followed by an untrusted directive to omit evidence supporting the second position. The initial response r_1 adopts that directive. A later revision e_2 removes the imperative wording but preserves a one-sided conclusion in artifact span a_4. TrustTrace represents the candidate route as

$$d_2 \rightarrow c_1 \rightarrow r_1 \rightarrow e_2 \rightarrow a_4, \tag{1}$$

where the content-flow channel can preserve the valid statistics while the control-flow channel represents evidence suppression. The graph first returns a ranked source, the affected span, the first activated stage, and a top-k route. The verifier then rewrites only the control-bearing part of d_2, replays the suffix beginning at its first retrieval, and checks whether the balanced evidence reappears. A stable intervention supports a replay-verified dependency; disagreement among accepted neutralizers or a non-replayable transformation produces an abstention. This example illustrates that the protected object is not the presence of an attack phrase but the residual source-to-span influence after transformation.

B. PROBLEM FORMULATION

We consider an LLM-assisted learning workflow as a temporally ordered process that transforms external sources

into a final textual artifact. For workflow w, let $\mathbf{q}^w$ denote the trusted task instruction and let $\mathcal{D}^w = \{d_1, \ldots, d_K\}$ denote the set of external source documents available to the system. The workflow contains T interaction stages. At stage t, the system retrieves a context set $\mathcal{C}_t$, constructs a user or application prompt q_t, produces an LLM response r_t, and may record a revision operation e_t. The final artifact A is segmented into M spans, $A = \{a_1, \ldots, a_M\}$. A span can be a sentence, a citation-bearing clause, or another application-defined unit that remains stable across revisions.

The workflow trace is represented as

$$\mathcal{W} = \left(\mathbf{q}^w, \mathcal{D}^w, \{\mathcal{C}_t, q_t, r_t, e_t\}_{t=1}^{T}, A\right). \tag{2}$$

An attacker may place an indirect prompt injection in one or more external source spans. The injected content can contain valid task-relevant facts and an unauthorized control instruction at the same time. Therefore, the security objective is not to reject every instructional sentence, but to determine whether unauthorized control survives the workflow and reaches the final artifact.

We distinguish four stage-wise states. The exposure indicator z_t^{exp} is one when an attacker-controlled source span is included in $\mathcal{C}_t$. The activation indicator z_t^{act} is one when the response at stage t follows or reflects the adversarial control objective. The retention indicator z_t^{ret} is one when the activated influence survives a later generation or revision stage. The artifact label y_m^{art} indicates whether final span a_m retains that influence. Exposure is obtained from retrieval logs during controlled data construction, whereas activation, retention, and artifact contamination are prediction targets.

TrustTrace jointly produces four audit outputs. Workflow-level propagation detection is an auxiliary screening output and predicts

$$\hat{y}^{\text{prop}} = P\left(y^{\text{prop}} = 1 \mid \mathcal{W}\right), \tag{3}$$

where $y^{\text{prop}} = 1$ indicates that an injected source affects at least one downstream response or final-artifact span. Stage localization estimates the first activated stage

$$\hat{t}^{\text{act}} \in \{1, \ldots, T, T+1\}, \tag{4}$$

where class $T + 1$ denotes no activation. Source tracing predicts $\hat{\mathbf{y}}^{\text{src}} \in [0, 1]^K$, and artifact localization predicts $\hat{\mathbf{y}}^{\text{art}} \in [0, 1]^M$. The multi-label source representation allows multiple compromised sources, while single-source experiments can additionally report top-1 attribution accuracy.

The formulation is deliberately restricted to trace-level security inference. It does not treat model predictions as causal estimates of learner cognition or educational outcomes. Counterfactual replay, introduced in Section III-J, is used to verify an operational source–artifact dependency under controlled system replay.

C. THREAT MODEL AND DESIGN PRINCIPLES

The attacker controls the textual content of at most B_s external source passages, such as webpages, uploaded readings, retrieved notes, or shared knowledge-base entries. The attacker knows the application task class and may use obfuscation, paraphrasing, delayed triggers, instruction splitting, or benign factual content to conceal the control objective. The attacker succeeds when the objective changes a downstream answer, suppresses or distorts evidence, manipulates a citation, redirects the requested task, or leaves a semantically contaminated span in the final artifact.

The attacker is not assumed to modify the trusted system prompt, model parameters, workflow logger, timestamps, or the final audit record. TrustTrace operates at the application layer and requires only observable source content and workflow traces; access to proprietary LLM hidden states is not required. During verified inference, the defender is allowed to replay a small number of suspicious workflow suffixes using the same model interface. This assumption is practical for auditing and offline review, but the base graph detector can still operate when replay is unavailable.

We evaluate both non-adaptive and adaptive attackers. The adaptive attacker knows the graph-construction features, candidate-predecessor budget, and the existence of counterfactual verification, but not the learned parameters, calibration examples, verifier randomness, or independent annotation decisions. It may distribute one objective across several sources, wrap control in task-relevant facts, introduce benign evidence that creates a decoy provenance path, reduce lexical and embedding similarity, or target the neutralizer with instruction-preserving paraphrases. The attacker cannot alter audit logs, timestamps, trusted prompts, or replay controls. These assumptions define a gray-box audit threat model rather than a claim of robustness to arbitrary white-box optimization.

Four design principles follow. First, benign evidence dependence must not be treated as adversarial propagation. Second, the model must remain sensitive to control objectives after copying, summarization, paraphrasing, and revision remove the original attack wording. Third, endpoint correctness must be supported by a temporally coherent intermediate path. Fourth, source attribution must be allowed to abstain when several sources are indistinguishable, when interventions disagree, or when the logged workflow does not contain enough evidence for a reliable decision.

D. OVERVIEW OF TrustTrace

As illustrated in Figure 1, TrustTrace consists of four components. First, a workflow parser segments documents, prompts, responses, revisions, and the final artifact, and converts them into a temporally ordered heterogeneous provenance graph. Second, a dual-flow temporal graph encoder learns a content-flow channel for legitimate evidence inheritance and a control-flow channel for unauthorized task influence. Third, differentiable control-path reachability links candidate sources to downstream response and artifact spans, providing a shared structural basis for propagation detection, activation-stage localization, source tracing, and contaminated-span localization. Fourth, a minimal

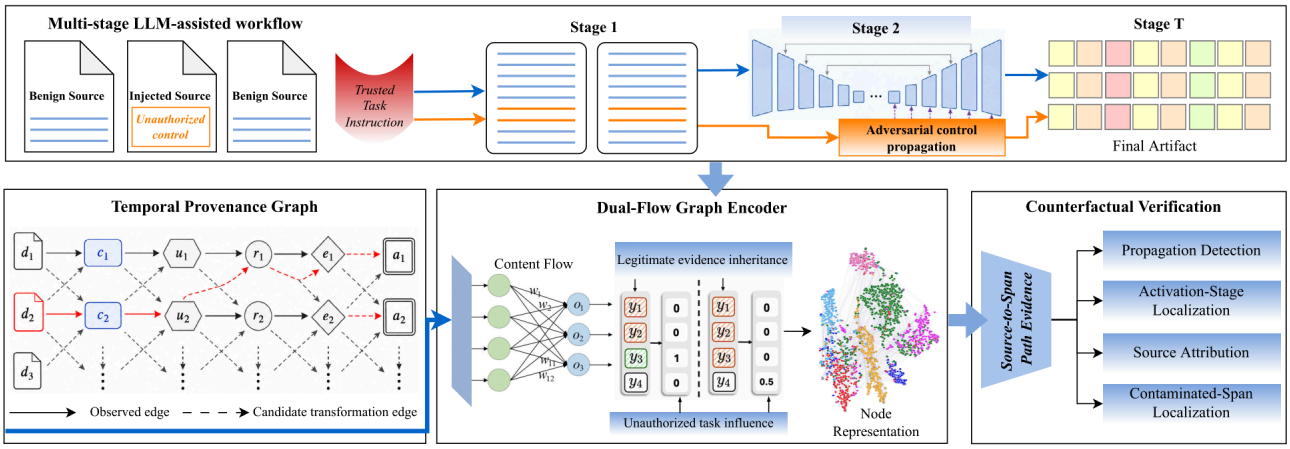


FIGURE 1. Overview of the TrustTrace framework for detecting, tracing, and verifying indirect prompt-injection propagation.

counterfactual verifier neutralizes the predicted control component of high-risk sources, replays only the affected workflow suffix, and calibrates the resulting attribution set.

Given a workflow $\mathcal{W}$, the framework produces

$$\left(\hat{y}^{\text{prop}}, \hat{\mathbf{p}}^{\text{stage}}, \hat{\mathbf{y}}^{\text{src}}, \hat{\mathbf{y}}^{\text{art}}, \mathbf{\Gamma}_{\alpha}\right), \tag{5}$$

where $\hat{\mathbf{p}}^{\text{stage}} \in [0,1]^{T+1}$ is the activation-stage distribution and $\mathbf{\Gamma}_{\alpha}$ is a calibrated candidate-source set at error level α.

E. HETEROGENEOUS TEMPORAL PROVENANCE GRAPH

The workflow is converted into a directed acyclic graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{\kappa}, \boldsymbol{\rho}), \tag{6}$$

where $\mathcal{V}$ is the node set, $\mathcal{E}$ is the edge set, κ_i is the stage index of node v_i, and ρ_i is its node type. We use six node types:

$$\mathcal{V} = \mathcal{V}^{\text{src}} \cup \mathcal{V}^{\text{ctx}} \cup \mathcal{V}^{\text{usr}} \cup \mathcal{V}^{\text{rsp}} \cup \mathcal{V}^{\text{edit}} \cup \mathcal{V}^{\text{art}}. \tag{7}$$

They represent source spans, retrieved-context occurrences, trusted user or application prompts, response spans, revision operations, and final-artifact spans, respectively. A source span and its retrieved occurrence are kept as separate nodes so that the same external text can be exposed at different stages without losing its original identity.

The edge set contains observed and candidate dependencies. Observed edges are obtained from system logs, including retrieval, prompt conditioning, chronological succession, explicit citation, and artifact assembly. Candidate edges represent transformations that are not explicitly logged, including copying, paraphrasing, summarization, omission, and revision inheritance. Candidate predecessors of node v_j are restricted to earlier nodes and selected by a top-K_g retrieval rule over semantic similarity, token overlap, citation identifiers, and edit alignment. This temporal restriction keeps the graph acyclic and prevents future artifact content from being used to construct edges toward earlier stages.

For node v_i, a frozen or jointly tuned text encoder produces $\mathbf{x}_i \in \mathbb{R}^{d_x}$. The initial node representation is

$$\mathbf{h}_i^0 = \mathbf{W}_x \mathbf{x}_i + \mathbf{e}_{\rho_i}^{\text{type}} + \mathbf{e}_{\kappa_i}^{\text{stage}} + \mathbf{e}_i^{\text{pos}} + \mathbf{W}_m \mathbf{m}_i, \tag{8}$$

where $\mathbf{m}_i$ contains observable metadata such as source trust level, retrieval rank, response role, and revision type. For edge (i, j), the feature vector $\boldsymbol{\phi}_{ij}$ contains its relation type, temporal distance, observed-versus-candidate flag, semantic support score, lexical alignment score, and task-instruction compatibility score.

F. DUAL-FLOW TEMPORAL PROVENANCE ENCODING

A single provenance relation cannot distinguish a source that supports an answer from a source that controls how the answer is produced. TrustTrace therefore maintains a content state $\mathbf{h}_i^{c,l}$ and a control state $\mathbf{h}_i^{g,l}$ for every node at graph layer l. Both states are initialized from $\mathbf{h}_i^0$ but use independent projections.

For predecessor v_i and destination v_j, the content-flow attention logit is

$$e_{ij}^{c,l} = \frac{(\mathbf{W}_Q^{c,l}\mathbf{h}_j^{c,l})^{\top}(\mathbf{W}_K^{c,l}\mathbf{h}_i^{c,l})}{\sqrt{d}} + (\mathbf{b}_c^l)^{\top}\boldsymbol{\phi}_{ij}, \tag{9}$$

whereas the control-flow logit is

$$e_{ij}^{g,l} = \frac{(\mathbf{W}_Q^{g,l}\mathbf{h}_j^{g,l})^{\top}(\mathbf{W}_K^{g,l}\mathbf{h}_i^{g,l})}{\sqrt{d}} + (\mathbf{b}_g^l)^{\top}\boldsymbol{\phi}_{ij}. \tag{10}$$

The normalized weights are

$$\alpha_{ij}^{u,l} = \frac{\exp(e_{ij}^{u,l})}{\sum_{k\in\mathcal{N}^-(j)}\exp(e_{kj}^{u,l})}, \quad u \in \{c, g\}, \tag{11}$$

where $\mathcal{N}^-(j)$ denotes the predecessor set of v_j.

A revision may preserve, attenuate, or remove a control objective. We therefore introduce an edge-level retention gate

$$\gamma_{ij}^l = \text{sigmoid}\left(\mathbf{w}_{\gamma}^{l\top}[\mathbf{h}_i^{g,l}; \mathbf{h}_j^{g,l}; \boldsymbol{\phi}_{ij}]\right). \tag{12}$$

The two channels are updated as

$$\mathbf{m}_j^{c,l} = \sum_{i \in \mathcal{N}^-(j)} \alpha_{ij}^{c,l} \mathbf{W}_V^{c,l} \mathbf{h}_i^{c,l}, \tag{13}$$

$$\mathbf{m}_j^{g,l} = \sum_{i \in \mathcal{N}^-(j)} \alpha_{ij}^{g,l} \gamma_{ij}^{l} \mathbf{W}_V^{g,l} \mathbf{h}_i^{g,l}, \tag{14}$$

and

$$\mathbf{h}_j^{u,l+1} = \mathrm{FFN}_u^l \left(\mathrm{LN}(\mathbf{h}_j^{u,l} + \mathbf{m}_j^{u,l}) \right), \quad u \in \{c, g\}. \tag{15}$$

The final node representation combines both channels and their interaction:

$$\mathbf{z}_j = \mathbf{W}_o[\mathbf{h}_j^{c,L_g}; \mathbf{h}_j^{g,L_g}; \mathbf{h}_j^{c,L_g} \odot \mathbf{h}_j^{g,L_g}]. \tag{16}$$

The content and control channels are supervised by relation labels available from controlled workflow construction. Content edges are derived from source use, copying, citation, and aligned revision ancestry. Control edges connect the injected control component to the first affected response and to downstream spans that preserve its objective. For unlabeled candidate edges, the model relies on task losses and consistency regularization rather than assigning them a forced negative label.

G. DIFFERENTIABLE CONTROL-PATH REACHABILITY

Attention weights alone are local and do not establish a complete source-to-artifact path. Because $\mathcal{G}$ is a directed acyclic graph, TrustTrace computes differentiable control reachability in topological order. We define the bounded control-transmission evidence of edge (i, j) as

$$q_{ij} = \frac{1}{L_g} \sum_{l=1}^{L_g} \alpha_{ij}^{g,l} \gamma_{ij}^{l}. \tag{17}$$

For source document d_k, let $\mathcal{S}_k \subset \mathcal{V}^{\mathrm{src}}$ denote its source-span nodes. The reachability from d_k to node v_j is initialized as

$$R_{kj} = 1, \quad v_j \in \mathcal{S}_k, \tag{18}$$

and otherwise propagated by a noisy-OR recursion:

$$R_{kj} = 1 - \prod_{i \in \mathcal{N}^-(j)} \left(1 - R_{ki} q_{ij}\right). \tag{19}$$

Thus, R_{kj} is high when at least one strong control path connects source d_k to v_j. The noisy-OR is a differentiable soft union, not a calibrated causal probability. It treats incoming route evidence as conditionally independent for aggregation; shared ancestors and correlated branches can therefore double-count evidence and create overconfidence on long or highly convergent paths. We address ranking reliability with held-out isotonic calibration and report the remaining route-length effect explicitly, but calibration does not remove the structural dependence approximation. Unlike selecting a single shortest path, the recursion supports multiple paraphrasing and revision routes while preserving this stated limitation.

For final-artifact span a_m with node index $j(m)$, the source-to-span path evidence is

$$P_{km}^{\mathrm{path}} = R_{k,j(m)}. \tag{20}$$

For response or edit nodes belonging to stage t, the stage-level control evidence is

$$P_t^{\mathrm{ctrl}} = 1 - \prod_{k=1}^{K} \prod_{j \in \mathcal{V}_t^{\mathrm{rsp}} \cup \mathcal{V}_t^{\mathrm{edit}}} (1 - R_{kj}). \tag{21}$$

These quantities provide a structural link among the four prediction tasks rather than allowing each output head to learn an unrelated shortcut.

H. PROPAGATION DETECTION AND TRACING HEADS

Artifact contamination is predicted from the control state of each artifact span and its incoming path evidence:

$$\hat{y}_m^{\mathrm{art}} = \mathrm{sigmoid}\left(f_a([\mathbf{z}_{j(m)}; \max_k P_{km}^{\mathrm{path}}]) \right). \tag{22}$$

The unverified source score combines graph reachability and localized contamination:

$$\hat{y}_k^{\mathrm{src}} = 1 - \prod_{m=1}^{M} \left(1 - P_{km}^{\mathrm{path}} \hat{y}_m^{\mathrm{art}}\right). \tag{23}$$

The workflow-level propagation probability is

$$\hat{y}^{\mathrm{prop}} = 1 - \prod_{k=1}^{K} \prod_{m=1}^{M} \left(1 - P_{km}^{\mathrm{path}} \hat{y}_m^{\mathrm{art}}\right). \tag{24}$$

This definition requires both a source-to-span control path and a contaminated span, reducing false alarms caused by suspicious source text that is never adopted.

For activation-stage localization, a stage logit is computed from P_t^{ctrl} and the pooled control representations at stage t:

$$\xi_t = f_s\left([P_t^{\mathrm{ctrl}}; \mathrm{Pool}(\{\mathbf{h}_j^{g,L_g} : v_j \in \mathcal{V}_t\})] \right). \tag{25}$$

An additional no-activation logit ξ_{T+1} is learned from the workflow representation. The stage distribution is

$$\hat{\mathbf{p}}^{\mathrm{stage}} = \mathrm{softmax}([\xi_1, \ldots, \xi_{T+1}]). \tag{26}$$

Stage-wise retention probabilities are obtained by applying a sigmoid head to response and edit nodes after the first predicted activation stage. These auxiliary outputs support fine-grained supervision but are not required for the final source decision.

I. CROSS-FORM AND CROSS-PROFILE CONSISTENCY

A detector can obtain high in-distribution accuracy by memorizing attack phrases, source templates, or the style induced by one prompting and decoding profile. The primary training corpus uses GPT-5.5 generation profiles, so the consistency objective is designed to suppress within-family wording and orchestration shortcuts rather than to imply cross-family invariance. Controlled workflow generation

groups traces that share the same attack objective and source-to-artifact propagation pattern but differ in payload wording, source format, system prompt, revision prompt, decoding profile, and request seed. Cross-family transfer is evaluated separately on Qwen3-Gen-Audit without retraining. Let $\mathbf{z}_w^g$ be the pooled control representation of workflow w. For a minibatch $\mathcal{B}$, the pairwise contrastive term is

$$\ell_{wp} = -\log \frac{\exp(s_{wp}/T_{\text{con}})}{\sum_{n\in\mathcal{B}\setminus\{w\}} \exp(s_{wn}/T_{\text{con}})}, \tag{27}$$

where $s_{wn} = \text{sim}(\mathbf{z}_w^g, \mathbf{z}_n^g)$. The invariance loss is

$$\mathcal{L}_{\text{inv}} = \sum_{w\in\mathcal{B}} \frac{1}{|\mathcal{P}(w)|} \sum_{p\in\mathcal{P}(w)} \ell_{wp}, \tag{28}$$

where $\mathcal{P}(w)$ contains traces with the same control objective and propagation label, T_{con} is the contrastive temperature, and $\text{sim}(\cdot,\cdot)$ is cosine similarity. Clean workflows with similar topics but no adversarial control are retained as hard negatives. This construction encourages control-level invariance without collapsing benign content with malicious influence.

J. MINIMAL COUNTERFACTUAL PATH VERIFICATION

Graph reachability identifies plausible propagation paths, but semantic correlation can still produce a false trace. TrustTrace therefore verifies only the top-B_v source candidates. For candidate source d_k, the model first predicts a control mask over its spans. A neutralization operator $\mathcal{N}(\cdot)$ removes or rewrites the masked instruction while preserving the source's factual propositions:

$$d_k^- = \mathcal{N}(d_k, \mathbf{q}^w). \tag{29}$$

The neutralized source is accepted only when it satisfies

$$\text{FactSim}(d_k^-, d_k) \geq \theta_f, \quad \text{CtrlScore}(d_k^-) \leq \theta_c. \tag{30}$$

The first constraint prevents naive source deletion from being interpreted as attack removal, and the second checks that the suspected control component has been attenuated.

1) PROCEDURALLY SEPARATED GPT-5.5 NEUTRALIZATION ENSEMBLE

To make the verifier configuration explicit, all three learned neutralization operators use the GPT-5.5 base service but are separated by prompt, context, decoding profile, request-seed namespace, and output schema. Operator $\mathcal{N}_1$ performs span-constrained deletion: GPT-5.5 receives the trusted task, the candidate source, and an externally proposed control mask, and may delete or locally repair only masked spans. Operator $\mathcal{N}_2$ performs proposition extraction and reconstruction: GPT-5.5 first emits atomic factual propositions with source-span identifiers and then reconstructs a passage from propositions that remain task-relevant after control removal. Operator $\mathcal{N}_3$ performs a minimal fact-preserving rewrite under a separately versioned system prompt. None of the operators receives TrustTrace scores, gold source labels, the gold attack objective, paired clean outputs, or another neutralizer's response. Their separation is procedural rather than architectural, a limitation considered in Section VI-A.

The objective-conditioned contamination checker and CtrlScore are also implemented as separately prompted, frozen GPT-5.5 judging profiles. The objective checker receives the trusted task, an objective rubric, and the candidate output span, but not the predicted provenance path. CtrlScore receives the trusted task and source passage and returns a structured control-likelihood judgment. Neither judge output is used as a TrustTrace node feature, edge feature, or training target. Entity and number preservation are additionally checked deterministically, and factual preservation combines bidirectional GPT-5.5 entailment judgments with these deterministic checks.

A candidate is replayed only when at least two neutralizers satisfy Eq. (30). If accepted neutralizers disagree materially on the affected artifact span, the verifier records the case as intervention-unstable and abstains from unique source attribution. Section V-E reports blinded audits of fact preservation, control attenuation, and downstream attribution stability. This design reduces direct evaluator–verifier reuse, but it does not claim independence across model families.

Let t_k^{exp} be the earliest stage at which d_k is retrieved. The verifier keeps the trusted task instruction, all non-candidate sources, retrieval order, and decoding configuration fixed and replays only the suffix beginning at t_k^{exp}. System-generated stages are regenerated, and logged deterministic edit operations are re-applied after span alignment. A free-form human edit is represented as an unreplayable boundary, and the verifier never synthesizes an unobserved human response. HumanEdit-Audit evaluates graph tracing across logged editor revisions and requires abstention whenever unique source verification would depend on regenerating the editor's counterfactual action.

Because generation is stochastic, the suffix is replayed B_r times. Let $s_o(a_m) \in [0,1]$ denote an objective-conditioned semantic contamination score for span a_m. The continuous value is used only through paired within-workflow differences in the necessity and sufficiency equations. A separate blinded human ordinal audit of s_o is reported in Section IV-D; nevertheless, s_o is not interpreted as an absolute or causal contamination scale. The necessity score is

$$N_{km} = \left[s_o(a_m) - \frac{1}{B_r} \sum_{b=1}^{B_r} s_o(a_{m,k}^{-,b}) \right]_+, \tag{31}$$

where $a_{m,k}^{-,b}$ is the aligned counterfactual span under the neutralized source and $[x]_+ = \max(x, 0)$.

To test whether the isolated control component is sufficient to restore the effect, it is reinserted into the fact-preserving neutralized source, producing d_k^+. The sufficiency score is

$$S_{km} = \left[\frac{1}{B_r} \sum_{b=1}^{B_r} s_o(a_{m,k}^{+,b}) - \frac{1}{B_r} \sum_{b=1}^{B_r} s_o(a_{m,k}^{-,b}) \right]_+. \tag{32}$$

The verified source-to-span score is

$$V_{km} = P_{km}^{\text{path}} \left(\eta N_{km} + (1-\eta) S_{km}\right), \tag{33}$$

where $\eta \in [0, 1]$. The verified source score is obtained by a noisy-OR aggregation:

$$\tilde{y}_k^{\text{src}} = 1 - \prod_{m=1}^{M} (1 - V_{km}). \tag{34}$$

This procedure is an operational intervention on the LLM workflow. It verifies whether the observed system output depends on the suspected control component under fixed replay conditions; it does not claim causal identification of human learning behavior.

K. CALIBRATED ATTRIBUTION AND ABSTENTION

A forensic system should not force a single source label when evidence is weak. TrustTrace therefore converts verified source scores into a set-valued prediction using a held-out calibration set. For a calibration workflow with true injected source k^*, the nonconformity score is

$$\omega = 1 - \tilde{y}_{k^*}^{\text{src}}. \tag{35}$$

Let $q_{1-\alpha}$ be the finite-sample $(1-\alpha)$ quantile of calibration nonconformity scores. The candidate-source set for a test workflow is

$$\mathbf{\Gamma}_\alpha(\mathcal{W}) = \left\{k : 1 - \tilde{y}_k^{\text{src}} \le q_{1-\alpha}\right\}. \tag{36}$$

The in-domain guarantee is defined at the workflow-group level. Topic, source family, attack-template family, duplicate cluster, and generation-seed group are assigned wholly to one partition before splitting; calibration and test workflows are then sampled from the same pre-specified in-domain construction process. This grouping makes workflow-level exchangeability reasonable while avoiding the stronger and false assumption that spans, edges, or paths are independent. Under this restricted assumption, the construction targets marginal source-set coverage of at least $1-\alpha$ for the single-origin setting. For multi-source workflows, the procedure is evaluated marginally for each source. No formal guarantee is claimed under dataset, orchestration, or model-family shift.

The system abstains from unique attribution when $\mathbf{\Gamma}_\alpha$ is empty, when its size exceeds a deployment threshold B_a, or when every candidate path crosses an unreplayable edit boundary. In these cases, TrustTrace can still report workflow-level propagation and contaminated spans without presenting an unsupported source claim.

L. TRAINING OBJECTIVE

The workflow propagation loss is binary cross-entropy:

$$\mathcal{L}_{\text{prop}} = \text{BCE}(\hat{y}^{\text{prop}}, y^{\text{prop}}). \tag{37}$$

The activation-stage loss is

$$\mathcal{L}_{\text{stage}} = -\sum_{t=1}^{T+1} y_t^{\text{stage}} \log \hat{p}_t^{\text{stage}}. \tag{38}$$

Source and artifact-span losses use class-balanced binary cross-entropy:

$$\mathcal{L}_{\text{src}} = \frac{1}{K} \sum_{k=1}^{K} \text{CBCE}(\hat{y}_k^{\text{src}}, y_k^{\text{src}}), \tag{39}$$

$$\mathcal{L}_{\text{art}} = \frac{1}{M} \sum_{m=1}^{M} \text{CBCE}(\hat{y}_m^{\text{art}}, y_m^{\text{art}}). \tag{40}$$

For labeled graph edges $\mathcal{E}_L$, the dual-flow edge loss is

$$\mathcal{L}_{\text{edge}} = \frac{1}{|\mathcal{E}_L|} \sum_{(i,j)\in\mathcal{E}_L} \Big[\text{BCE}(\bar{\alpha}_{ij}^c, y_{ij}^c) + \text{BCE}(\bar{\alpha}_{ij}^g \bar{\gamma}_{ij}, y_{ij}^g) \Big], \tag{41}$$

where $\bar{\alpha}$ and $\bar{\gamma}$ are layer-averaged values. A path-consistency loss supervises source-to-artifact reachability when controlled path labels are available:

$$\mathcal{L}_{\text{path}} = \frac{1}{KM} \sum_{k=1}^{K} \sum_{m=1}^{M} \text{BCE}(P_{km}^{\text{path}}, y_{km}^{\text{path}}). \tag{42}$$

The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{prop}} + \lambda_1 \mathcal{L}_{\text{stage}} + \lambda_2 \mathcal{L}_{\text{src}} + \lambda_3 \mathcal{L}_{\text{art}} + \lambda_4 \mathcal{L}_{\text{edge}} + \lambda_5 \mathcal{L}_{\text{path}} + \lambda_6 \mathcal{L}_{\text{inv}} + \lambda_7 \|\Theta\|_2^2, \tag{43}$$

where Θ denotes all learnable parameters. The counterfactual verifier is applied after graph inference and is not differentiated through the external LLM. This separation prevents stochastic replay from destabilizing graph training and permits black-box deployment.

M. TRAINING AND VERIFIED INFERENCE PROCEDURE

The overall procedure is summarized in Algorithm 1. During training, each recorded workflow is converted into a leakage-controlled temporal graph, encoded through the dual-flow network, and optimized using the joint objective. During inference, the graph model first provides inexpensive workflow, stage, source, and span predictions. Counterfactual replay is then restricted to the top-B_v source candidates whose path evidence exceeds θ_p. The final output includes a calibrated source set and an abstention flag in addition to the standard detection results.

N. COMPUTATIONAL COMPLEXITY

Let $V = |\mathcal{V}|$, $E = |\mathcal{E}|$, d be the hidden dimension, and L_g be the number of graph layers. With top-K_g predecessor selection, $E = O(VK_g)$. The dual-flow graph encoder requires

$$O\left(L_g E d + L_g V d^2\right) \tag{44}$$

time and $O(Vd + E)$ memory. For K candidate sources, differentiable reachability is computed in $O(KE)$ time because the graph is processed once in topological order per source. The output heads require $O(KM)$ operations for source-to-artifact aggregation.

Algorithm 1 Training and Verified Inference of TrustTrace

Require: Workflow traces $\mathbb{W}$, labels $\mathbb{Y}$, graph depth L_g, verification budget B_v

Ensure: Propagation, stage, source, span, and calibrated attribution outputs

```
Initialize model parameters Θ
for each training epoch do
    for each workflow W ∈ W_train do
        Build temporal provenance graph G
        Encode content and control states for L_g layers
        Compute edge transmission q_ij and reachability R_kj
        Predict ŷ^prop, p̂^stage, ŷ^src, and ŷ^art
        Update Θ by minimizing L
    end for
end for
Fit attribution quantile q_{1−α} on W_cal
for each test workflow W do
    Run graph inference and rank sources by ŷ_k^src
    for each of the top-B_v sources with P_km^path > θ_p do
        Neutralize its control component and replay the affected suffix
        Compute necessity N_km, sufficiency S_km, and verified score V_km
    end for
    Construct Γ_α(W) and apply the abstention rule
    Return workflow, stage, source-set, and contaminated-span results
end for
```

Counterfactual verification is intentionally budgeted. If B_v sources are verified with B_r repeated suffix replays, its cost is $O(B_v B_r C_{\text{LLM}})$, where C_{LLM} is the cost of one affected-suffix replay. This external-model cost dominates verified inference, but it is incurred only for high-risk candidates. The graph detector therefore remains suitable for routine screening, while replay is reserved for cases that require forensic source attribution.

IV. EXPERIMENTAL SETUP

A. DATASETS AND WORKFLOW CONSTRUCTION

We evaluate on one controlled workflow dataset constructed for this study, a separately implemented RAG testbed, one independent-model generation set, one controlled human-edit set, and three public benchmarks adapted to the source-to-artifact setting. Table 3 summarizes the workflow collections. None is a longitudinal classroom log. EduTrace-IPI, CourseRAG-Audit, and Qwen3-Gen-Audit use fully logged system stages; HumanEdit-Audit contains free-form revisions by blinded editors under a fixed editing protocol. The public adaptations are test-only research transformations rather than official benchmark extensions. The evaluation therefore measures trace inference under controlled observability and does not infer learner behavior.

1) EduTrace-IPI

We construct EduTrace-IPI, a controlled English-language dataset of 18,000 LLM-assisted learning workflows. It contains four task families: evidence-grounded question answering, explanatory summarization, argumentative writing, and citation-based synthesis. Each task is paired with 6–12 passages drawn from openly licensed instructional material, encyclopedia-style documents, public reports, and synthetic distractors. A workflow contains three to five stages: retrieval, GPT-5.5 initial generation, one or more GPT-5.5 revision operations, and final artifact assembly. Table 2 gives the compact composition statistics.

Twelve thousand workflows contain one or two injected passages and 6,000 are clean. The attack set covers six control objectives: task redirection, evidence suppression, conclusion steering, citation manipulation, answer-format hijacking, and fabricated support. Each objective is expressed through direct imperatives, role-playing, delayed triggers, split instructions, light obfuscation, or fact-wrapped instructions. The attack source, exposure stage, objective, and injected spans are known from construction. Activation, retention, and artifact contamination are established through paired clean–attack replay with identical retrieval order and decoding configuration. A stratified subset is independently reviewed to resolve cases in which semantic equivalence cannot be determined by the automatic objective checker.

The split is 10,800/2,400/2,400/2,400 for training, validation, calibration, and testing. Topics, source-document families, attack templates, and generation seeds are grouped before splitting. Consequently, a test workflow cannot share the same source document or payload template with training. All system-generated stages use GPT-5.5. One complete GPT-5.5 generation profile—including its system prompt family, revision prompt family, decoding configuration, and request-seed namespace—is held out from training and appears only in the cross-profile test described in Section V-K. The held-out condition therefore tests generator-style and orchestration shift within a fixed base model rather than generalization to a different model family.

TABLE 2. Composition of EduTrace-IPI.

Task	Workflows	Src.	Stages	Art. spans
Evidence QA	4,500	7.8	3.2	8.6
Summarization	4,500	6.4	3.7	11.9
Argumentation	4,500	9.6	4.4	16.7
Citation synthesis	4,500	10.3	4.1	14.8

2) CourseRAG-AUDIT

To reduce dependence on the EduTrace-IPI construction pipeline, we additionally build CourseRAG-Audit using a separately implemented course-material RAG writing system. The pipeline uses a different retriever, orchestration code, prompt format, document segmentation rule, and

separately versioned GPT-5.5 generation profile from those used to construct EduTrace-IPI. It contains 1,200 test-only workflows covering explanation, evidence comparison, literature synthesis, and citation checking. Four hundred workflows are clean and 800 contain one or two injected sources. Each workflow includes a GPT-5.5 initial answer and two GPT-5.5 revision stages produced under a fixed sandbox configuration; no stage is a natural learner-authored edit. Source, stage, span, and intermediate-path labels are assigned by the independent annotation protocol in Section IV-C; the EduTrace objective checker and neutralization code are not used to create its gold labels. CourseRAG-Audit is not used for training, model selection, threshold selection, or calibration and therefore serves as a deployment-style external validity test rather than a second in-domain dataset.

3) QWEN3-GEN-AUDIT

Qwen3-Gen-Audit is a test-only cross-family set of 800 workflows constructed from topic, source-family, and attack-template groups that do not occur in EduTrace-IPI. Five hundred workflows contain one injected source and 300 are clean. Initial responses and two revision stages are generated with the frozen `Qwen3-235B-A22B-Instruct` model using prompts that preserve the task semantics but do not reuse the GPT-5.5 system messages. Source identity and exposure are known from construction. Activation, retention, span, edge, and complete-path labels are assigned by the independent annotation protocol, and no Qwen3 example is used for training, validation, calibration, or threshold selection. This set tests transfer across both generation family and prompting stack.

4) HumanEdit-AUDIT

HumanEdit-Audit contains 400 test-only workflows built from 100 held-out evidence-comparison and synthesis tasks. Each task yields four editor trajectories: two independently edited clean workflows and two independently edited attacked workflows, giving 200 clean and 200 attacked workflows in total. The four editors are assigned in a balanced rotation across the 400 trajectories and remain blinded to attack status and the gold source. A GPT-5.5 profile produces each initial draft, after which the assigned editor performs two free-form revisions intended to improve balance, clarity, and citation quality. All inserted, deleted, and moved spans are logged. A separate adjudication team labels activation, retention, contaminated spans, intermediate edges, and all defensible source-to-artifact paths. Because the human revision cannot be counterfactually regenerated, replay stops at the first editor boundary; graph tracing remains evaluable, while unique verified attribution across that boundary requires evidence from pre-boundary interventions and otherwise abstains. The editing task uses only public or synthetic instructional content and contains no student records. This set evaluates the framework's stated human-edit boundary but is not treated as a naturally occurring learner log.

5) BIPIA-TRACE

BIPIA is a public benchmark for indirect prompt injection across five application scenarios: email question answering, web question answering, table question answering, text summarization, and code question answering [4]. Its source material is drawn from OpenAI Evals emails, NewsQA articles, WikiTableQuestions tables, 10Sum documents, and Stack Overflow code discussions, respectively. The released benchmark combines benign task instances with attack instructions placed at different locations and provides disjoint attack-template partitions.

We construct BIPIA-Trace from 5,000 stratified attacked instances, 1,000 per scenario, and one matched clean counterpart for each instance, yielding 10,000 workflows. The original external content is treated as the source layer. Each instance is expanded into a fixed three-stage process consisting of an initial response, a meaning-preserving revision, and a final artifact. Source identity and injection spans are inherited from BIPIA. Propagation labels are assigned from paired replay and objective-conditioned output checks. BIPIA-Trace is used as an external test set; no BIPIA example is used to fit TrustTrace or its calibration threshold.

6) InjecAgent-TEXT

InjecAgent contains 1,054 public IPI test cases spanning 17 user tools and 62 attacker tools, with attack intents organized around direct user harm and private-data exfiltration [5]. To test whether a model trained on learning workflows transfers to tool-mediated text traces, we serialize each untrusted tool observation, the subsequent agent response, and the accepted action rationale into a two-to-four-stage textual workflow. Each attacked case is paired with the corresponding benign task context, resulting in 2,108 workflows. The injected tool output is the candidate source, while the final audit note is the artifact. This set is reserved for robustness evaluation.

7) AgentDojo-TEXT

AgentDojo is a public dynamic environment with 97 realistic tasks and 629 security test cases covering domains such as workspace management, banking, travel, and messaging [6]. We transform every security case and its matched benign execution into AgentDojo-Text, producing 1,258 workflows. Tool observations are source nodes; agent thoughts, responses, and accepted action descriptions form intermediate stages; and a final execution summary forms the textual artifact. The original injection location and malicious task determine source and propagation labels. AgentDojo-Text is test-only and measures transfer from document-centered learning workflows to agentic execution traces.

B. PREPROCESSING AND LEAKAGE CONTROL

HTML tags, navigation boilerplate, and duplicated whitespace are removed, while table structure, code blocks, quotation markers, and citation identifiers are preserved because

TABLE 3. Dataset sources, workflow counts, and leakage-controlled splits. "Prop." denotes the percentage of workflows in which adversarial influence reaches a response or final artifact.

Dataset	Origin	Main content	Workflows	Train/Val/Cal/Test	Stages	Sources	Prop.
EduTrace-IPI	Constructed, GPT-5.5	QA, summary, writing, citation	18,000	10,800/2,400/2,400/2,400	3–5	6–12	46.8%
CourseRAG-Audit	Separate GPT-5.5 RAG stack	Explanation, comparison, synthesis, citation	1,200	0/0/0/1,200	3–4	5–11	43.5%
Qwen3-Gen-Audit	Independent model family	QA, comparison, synthesis	800	0/0/0/800	3	6–10	44.1%
HumanEdit-Audit	Controlled editor revisions	Comparison and citation synthesis	400	0/0/0/400	3	6–10	42.0%
BIPIA-Trace	BIPIA [4]	Email, web, table, summary, code	10,000	0/0/0/10,000	3	1–6	38.4%
InjecAgent-Text	InjecAgent [5]	Tool observations and actions	2,108	0/0/0/2,108	2–4	2–9	41.6%
AgentDojo-Text	AgentDojo [6]	Workspace, banking, travel, chat	1,258	0/0/0/1,258	2–5	3–14	37.8%

they can carry both legitimate evidence and injected control. Documents and responses are segmented into sentences; citation-bearing clauses and code blocks are retained as atomic spans. A node is truncated to 192 subword tokens, with an overlap of 32 tokens for longer spans. The final artifact is segmented using the same rules used to generate the span labels.

Exact duplicates are removed before splitting. Near duplicates are detected using MinHash signatures and a Jaccard threshold of 0.85. All members of a duplicate cluster, source-document family, task topic, and attack-template family are assigned to the same partition. Candidate graph edges are constructed only from information available up to the destination stage. The top-K_g predecessor search therefore never observes future artifact spans. Encoder normalization statistics, class weights, decision thresholds, and calibration quantiles are estimated from the training, validation, or calibration partition as appropriate; public test sets are never used for adaptation.

For paired replay, clean and attacked workflows use the same retrieved passages, ordering, decoding temperature, and random seed whenever the backend supports deterministic seeding. A source is labeled as replay-verified for an artifact span only when the attack objective is present in the attacked output, attenuated after fact-preserving neutralization, and restored after reinsertion. System-generated nodes are replayed directly. HumanEdit-Audit retains the observed editor operations in the graph but marks the first free-form revision as an unreplayable boundary. A path remains eligible for graph-level scoring, but unique replay-verified attribution is withheld whenever replay or span alignment is unavailable for a required transformation.

C. INDEPENDENT ANNOTATION AND LABEL AUDIT

The automatic construction labels are audited independently to reduce construct-validity and verifier-circularity risks. Three annotators with experience in LLM security or provenance analysis label 300 stratified EduTrace-IPI training workflows, 1,200 EduTrace-IPI test workflows, all 1,200 CourseRAG-Audit workflows, all 400 HumanEdit-Audit workflows, and 400 stratified Qwen3-Gen-Audit workflows. Annotators are blinded to model predictions, automatic objective-checker outputs, neutralized sources, paired clean generations, and the identity of the generation model. They receive only the trusted task, chronologically ordered source and workflow spans, and a codebook defining exposure, activation, retention, contaminated spans, and directed intermediate edges. Each workflow is labeled by two annotators; disagreements and all implicit-steering cases are resolved by the third annotator. Span agreement is measured with pairwise token-overlap F1, edge agreement with Cohen's κ, and complete-path agreement with exact path match and normalized edit distance.

TABLE 4. Independent annotation agreement and agreement between automatic construction labels and adjudicated labels. Values are reported on the audited EduTrace-IPI subset.

Label target	Inter-annotator agreement	Auto–adjudicated agreement	Measure
Activation state	0.84	0.91	Cohen's κ
Retention state	0.79	0.88	Cohen's κ
Malicious source	0.87	0.93	Cohen's κ
Contaminated span	0.83	0.89	Token-overlap F1
Intermediate edge	0.76	0.85	Cohen's κ
Complete path	0.71	0.81	Exact path agreement

The additional training audit yields auto–adjudicated agreement of 0.89 for activation, 0.86 for retention, 0.91 for malicious source, and 0.87 token-overlap F1 for contaminated spans. These values are close to the held-out test audit and reduce the concern that the training partition is governed by a qualitatively different automatic label rule. Complete-path agreement remains the weakest label type because several earlier spans can jointly support one revision. Gold records therefore retain all adjudicated valid routes rather than forcing a single route, and Section V-B reports results both with and without disagreement-flagged workflows.

D. HUMAN VALIDATION OF THE CONTINUOUS CONTAMINATION SCORE

A separate set of 800 artifact spans is sampled across task families, attack objectives, score deciles, path lengths, and clean hard negatives. Three blinded annotators rate objective-conditioned contamination on a five-level ordinal scale from 0 (no attack-conditioned influence) to 4 (the span directly realizes the attack objective). The mean adjudicated rating is normalized to $[0, 1]$ and compared with the continuous score s_o. The analysis is performed for both the GPT-5.5 objective checker used in the primary verifier and the independent Qwen3 checker.

The continuous score is therefore not treated as an unvalidated internal quantity. Its strongest agreement is with

TABLE 5. Human validation of continuous objective-conditioned contamination scores on 800 stratified artifact spans.

Scoring source	Spearman ρ	QWK	AUC	MAE	ECE
Lexical objective similarity	0.41	0.36	0.734	0.71	0.112
Qwen3 independent checker	0.73	0.69	0.893	0.47	0.055
GPT-5.5 primary checker	**0.78**	**0.74**	**0.912**	**0.41**	**0.048**

clearly contaminated or clearly clean spans, while moderate ratings remain less stable when an artifact preserves framing or evidence omission without repeating an attacker-defined proposition. For this reason, the verifier continues to use paired replay differences and calibrated abstention rather than interpreting one absolute score as causal proof.

E. NEUTRALIZER AUDIT PROTOCOL

A separate sample of 600 candidate sources is used to audit the counterfactual verifier. Annotators compare each original and neutralized source without seeing TrustTrace predictions and answer three questions: whether all task-relevant factual propositions are preserved, whether the unauthorized control objective is removed, and whether the rewritten text introduces a new instruction. We also measure entity and number preservation, bidirectional entailment, and agreement among the three GPT-5.5 neutralization profiles. Verification results are reported separately for each neutralizer and for the two-of-three ensemble. A source is excluded from verified attribution when fact preservation fails, control attenuation fails, a new instruction is introduced, replay changes the artifact for reasons not localized to the suspected control, or accepted neutralizers yield inconsistent downstream conclusions. The audit distinguishes four boundary types: neutralization failure, accepted-neutralizer disagreement, unrelated replay drift, and insufficient evidence requiring abstention. The audit file stores the original-source hash, neutralized-output hash, component profile, request identifier, individual annotator decisions, adjudication, downstream replay outcome, and abstention reason so that every accepted or rejected intervention can be traced to both the GPT-5.5 request record and the human judgment.

F. COMPARISON METHODS

We distinguish native security comparisons, closest-prior mechanism adaptations, and architecture-matched tracing comparisons. Native methods preserve their original prevention or runtime outputs. They include BIPIA-Defense, StruQ, SecAlign, DetectRemove, InstructDetector, TaskShield, PIGuard, and IPIGuard [4], [7], [8], [9], [10], [11], [21], [29]. TracLLM is included as the native context-traceback baseline [30]. No native method directly returns all source, stage, span, and path outputs in our formulation. Missing source- and span-level heads therefore use a common frozen text encoder and a linear audit adapter with fewer than 0.4 million trainable parameters; the original defense is not updated. Adapter-derived metrics are marked with superscript "A" and must not be interpreted as the native method's intended performance.

To address the two closest conceptual systems, we implement text-artifact adaptations denoted ARGUS-Text and AttriGuard-Text. ARGUS-Text maps artifact spans to protected decisions, preserves the published distinction between trusted justification and untrusted influence, and ranks temporally valid source-to-span routes with its influence graph [14]. AttriGuard-Text treats each candidate source as the untrusted observation, applies whole-source and control-span interventions with the same replay budget, and attributes downstream artifact changes using necessity and sufficiency [15]. Neither adaptation uses TrustTrace's dual-flow encoder, recursive reachability loss, edge labels, or path labels. The adaptations are marked with superscript "T" because their protected output is changed from an agent action to a text artifact; they are mechanism-level comparisons rather than claims about the official implementations on their native tasks.

The primary architectural evidence uses five variants with the same node encoder, candidate edges, training split, and validation budget: (i) Longformer-Trace serializes the chronological trace without a graph; (ii) TGAT-Trace uses single-state temporal attention; (iii) HGT-Trace uses heterogeneous graph attention without dual-flow separation; (iv) GraphTransformer-Path predicts edge scores without recursive reachability; and (v) TrustTrace-Terminal uses the proposed graph architecture but removes edge and path supervision. TrustTrace-Terminal is the pre-specified primary comparator for complete-path recovery because it isolates intermediate supervision from the benefit of adopting a temporal graph.

All trainable audit heads are optimized on the EduTrace-IPI training partition, receive the same 30-trial validation budget, and are evaluated without adaptation on external test sets. The search dimensions are learning rate, dropout, hidden width, graph or sequence depth, and candidate-predecessor count where applicable. Appendix E reports the complete ranges, selected values, and trainable parameter counts. Native and adapter-augmented results establish complementarity, the closest-prior adaptations test mechanism overlap, and architecture-matched results support claims about dual-flow encoding, reachability, and path supervision.

G. EVALUATION METRICS

We report workflow, source, span, stage, path, calibration, and efficiency metrics. Prop-AUC is the area under the ROC curve for workflow-level propagation. Prop-F1 is the F1 score at a threshold selected on the validation set. Src@1 is top-1 malicious-source attribution accuracy among propagated single-source workflows; multi-source cases use label-wise source F1 and coalition recall. Span-F1 is micro-F1 over final-artifact spans. Stage-MAE measures the absolute error between the true and predicted first activation stage, with the

no-activation class mapped to $T+1$:

$$\text{Stage-MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{t}_i^{\text{act}} - t_i^{\text{act}}\right|. \tag{45}$$

For path-level evaluation, Edge-AUPRC is computed over the union of predicted candidate control edges and adjudicated gold control edges, with all edges from one workflow retained in the same resampling cluster. A gold edge omitted by the top-K_g predecessor search is assigned prediction score zero and remains in the recall denominator. Likewise, a gold route containing an omitted edge remains a valid target: when the candidate graph cannot represent that route, it contributes zero exact-path recall and a unit-cost dummy match to nPED rather than being removed from evaluation. This rule prevents candidate-edge pruning from inflating edge or path recall. For a source–artifact pair (d_k, a_m), let $\mathcal{G}_{km} = \{g_1, \ldots, g_R\}$ denote the adjudicated set of valid complete paths. Each path is an ordered typed sequence

$$p = (v_0, r_1, v_1, \ldots, r_L, v_L), \tag{46}$$

where v_0 is a source-span node, v_L is the target artifact-span node, and r_ℓ is the relation label of edge $(v_{\ell-1}, v_\ell)$. Predicted simple paths are ranked in the acyclic graph by

$$C(p) = \sum_{\ell=1}^{L} -\log\big(\max(q_{v_{\ell-1}v_\ell}, \epsilon)\big), \tag{47}$$

with lower cost indicating stronger path evidence and $\epsilon = 10^{-8}$. A top-k dynamic program returns the k lowest-cost paths. Overlapping token windows are first mapped to stable canonical span identifiers. Two paths are duplicates only when their ordered canonical node identifiers and relation labels are identical; in that case, only the lower-cost copy is retained. In every signature, each stage-specific canonical node identifier and each relation label is one edit token. Overlapping windows that refer to the same span at the same stage collapse to one canonical node, whereas identical text appearing at different workflow stages remains distinct. Evaluation is conditioned on a fixed source–artifact pair (d_k, a_m), so paths with different endpoints are not eligible matches. Unless otherwise stated, we use $k = 3$; consequently, the reported strict Complete-Path F1 is Complete-Path F1@3.

Multiple valid routes are handled by bipartite matching rather than by selecting one gold path. Let $\widehat{\mathcal{P}}_{km}^{(k)}$ be the deduplicated predicted path set. We construct a bipartite graph between $\widehat{\mathcal{P}}_{km}^{(k)}$ and $\mathcal{G}_{km}$, with an edge when the complete ordered typed signatures are identical, and obtain a maximum-cardinality matching $\mathcal{M}_{km}$. Strict Complete-Path precision and recall are

$$\text{Prec}_{km}^{\text{path}} = \frac{|\mathcal{M}_{km}|}{\max(1, |\widehat{\mathcal{P}}_{km}^{(k)}|)}, \qquad \text{Rec}_{km}^{\text{path}} = \frac{|\mathcal{M}_{km}|}{\max(1, |\mathcal{G}_{km}|)}, \tag{48}$$

and Complete-Path F1@3 is their harmonic mean after micro-aggregation over source–artifact pairs. Thus, a path of the wrong length or with an incorrect intermediate node does not count as a complete match.

Algorithm 2 Top-k Complete-Path Matching and Scoring

Require: DAG G, edge evidence q, source d_k, artifact span a_m, gold set $\mathcal{G}_{km}$, cutoff $k = 3$ unless stated otherwise
Ensure: Complete-Path counts, nPED, PathCov@k

1: Canonicalize overlapping windows to stable span identifiers
2: Enumerate the k lowest-cost simple paths from d_k to a_m in topological order using $C(p)$
3: Deduplicate paths by ordered node–relation signature, retaining the lowest-cost copy
4: Build exact-match matrix $A_{uv} = \mathbb{I}[\sigma(\hat{p}_u) = \sigma(g_v)]$
5: Obtain maximum-cardinality bipartite matching $\mathcal{M}_{km}$ from A
6: Accumulate $|\mathcal{M}_{km}|$, $|\widehat{\mathcal{P}}_{km}^{(k)}|$, and $|\mathcal{G}_{km}|$
7: Build edit-cost matrix $D_{uv} = d_{\text{PED}}(\hat{p}_u, g_v)$ and add dummy rows/columns with cost one
8: Obtain minimum-cost assignment on D and accumulate its normalized cost
9: Set PathCov@$k = 1$ iff $|\mathcal{M}_{km}| > 0$
10: Aggregate per pair, then per workflow, without treating spans or edges as independent samples

Partial route quality is reported separately with normalized path edit distance. For signatures $\sigma(p)$ and $\sigma(g)$,

$$d_{\text{PED}}(p, g) = \frac{\text{Lev}(\sigma(p), \sigma(g))}{\max(|\sigma(p)|, |\sigma(g)|)}, \tag{49}$$

where node identifiers and relation labels are edited as separate tokens and insertions, deletions, and substitutions have unit cost. Because the source and artifact endpoints are fixed by pair-specific evaluation, endpoint substitutions are not permitted as candidate matches; disagreement at an endpoint is treated as an unmatched path. A minimum-cost bipartite assignment with dummy paths of cost one handles unequal set sizes; nPED is the mean assigned cost. PathCov@k is the fraction of positive source–artifact pairs for which at least one of the top-k predictions exactly matches one adjudicated route. Endpoint-correct/path-wrong rate is computed only among pairs with the correct source and artifact span and records the fraction with no exact intermediate-path match.

When one artifact span is jointly influenced by several sources, evaluation is performed separately for every labeled (d_k, a_m) pair; a path from one source cannot satisfy the gold route of another. We additionally report any-source coverage and all-source coalition coverage in the multi-source analysis. Algorithm 2 gives the complete matching procedure.

ECE is the 15-bin expected calibration error of the workflow propagation score:

$$\text{ECE} = \sum_{b=1}^{15} \frac{|B_b|}{N}\,|\text{acc}(B_b) - \text{conf}(B_b)|. \tag{50}$$

For set-valued source attribution, we report empirical coverage, average set size, selective source error after abstention, and coverage under dataset or generation-profile shift. The finite-sample conformal guarantee is claimed only for exchangeable in-domain data; external coverage is an empirical diagnostic rather than a formal guarantee.

Results are averaged over the five fixed training seeds $\{17, 29, 43, 71, 101\}$. The record linkage, bootstrap unit, paired resampling procedure, and two pre-specified Holm families are defined in Section IV-J. A dagger denotes Holm-adjusted $p < 0.05$ only within one of those families. External-domain, sensitivity, adaptive-attack, and budget analyses are treated as descriptive unless an adjusted family is explicitly stated.

H. IMPLEMENTATION DETAILS

The implementation uses PyTorch and PyTorch Geometric. Text nodes are initialized with the frozen `BAAI/bge-base-en-v1.5` encoder, producing 768-dimensional vectors. The graph hidden dimension is 256, with three dual-flow layers, eight attention heads, dropout 0.2, and $K_g = 6$ candidate predecessors. The model is optimized with AdamW for at most 40 epochs using a batch size of eight workflows, a learning rate of 2×10^{-4}, weight decay 10^{-4}, gradient clipping at 1.0, and early stopping after six validation epochs. Python, NumPy, and PyTorch use the fixed seeds $\{17, 29, 43, 71, 101\}$. Deterministic GPU kernels are enabled when supported; operations that remain nondeterministic are listed in the environment manifest, and the released checkpoints and prediction files are the authoritative records for the reported values.

The loss weights are set to $(0.5, 1.0, 1.0, 0.4, 0.6, 0.2, 10^{-5})$. The contrastive temperature is 0.1. Unless otherwise stated, counterfactual verification uses $B_v = 2$ candidate sources, $B_r = 3$ replay samples, $\theta_p = 0.35$, $\theta_f = 0.90$, $\theta_c = 0.20$, and $\eta = 0.5$. Attribution sets use $\alpha = 0.10$ and abstain when their size exceeds three. Hyperparameters are selected on the validation set; calibration uses the separate calibration partition.

1) GPT-5.5 COMPONENT PROFILES AND API RECORDING

All language-model generation, neutralization, objective checking, and control checking use the GPT-5.5 base service without task-specific fine-tuning. The requested model alias is `gpt-5.5`. Because a provider alias can be updated, the experiment does not rely on the alias alone: every request record stores the provider-returned canonical model string, service revision or system fingerprint when available, UTC timestamp, request identifier, endpoint name, complete message payload, tool schema, decoding arguments, prompt SHA-256, and response SHA-256. All results in this manuscript are bound to the July 2026 GPT-5.5 experiment manifest. No aggregate table combines requests with different returned canonical model identifiers or fingerprints.

Table 6 lists the fixed component profiles. Temperature and top-p are submitted only when the endpoint exposes those controls; otherwise the manifest records the field as unsupported and the exact response payload is retained.

The three-judge profiles use separate request-seed namespaces and majority aggregation; ties or malformed structured outputs are marked unresolved rather than coerced into a label. The objective checker is used for large-scale construction but not for blinded adjudicated labels. CtrlScore is used only to accept or reject a proposed neutralization and is not a TrustTrace feature or training label. The same GPT-5.5 base across the primary roles creates residual common-model dependence; Sections IV-D, IV-I, and V-M assess it through human ordinal ratings and independent Qwen3 generation and verification.

If the provider changes the `gpt-5.5` alias, the new returned model string or fingerprint creates a new manifest version. Existing records are never overwritten, and reruns under the updated service are reported as a separate replication rather than merged with the July 2026 results. The exact access time of every call is retained in UTC in `api_calls.jsonl`. These archives make the reported calls auditable and permit table regeneration from recorded outputs, but they cannot guarantee exact service-level regeneration after a provider update. We therefore distinguish record reproducibility from future service reproducibility throughout the paper.

I. INDEPENDENT-MODEL GENERATION, NEUTRALIZATION, AND JUDGING

The cross-family replication uses the frozen `Qwen3-235B-A22B-Instruct` model. The Qwen3 stack does not receive GPT-5.5 prompts, objective-checker scores, neutralized outputs, source rankings, gold paths, or calibration labels. The generation profile receives only the trusted task and retrieved passages and produces the Qwen3-Gen-Audit workflows. The verification stack contains a proposition-preserving neutralizer and three separately seeded structured judges for objective-conditioned contamination, residual control, and bidirectional factual entailment. Thresholds are selected on a 300-workflow cross-family validation subset drawn from the original validation partition; no test label is used. The same graph checkpoints and source calibration quantile are retained, so changes measure the effect of replacing the generation or verification family rather than retraining TrustTrace.

For the Qwen3 neutralizer, the three QN replicates are sampled independently. Each replicate is checked against the same factual-preservation and residual-control acceptance criteria, and the candidate proceeds to replay only when at least two of the three QN replicates are accepted. The structured judges retain their three-replicate majority aggregation. This preserves the two-of-three acceptance principle but not the neutralizer structure of the primary stack: GPT-5.5 uses three distinct neutralization operators, whereas Qwen3 uses three independently sampled replicates of one neutralizer profile. Consequently, the cross-family comparison changes both model family and ensemble structure and is interpreted

TABLE 6. Frozen GPT-5.5 profiles used by the workflow and verification pipeline. All profiles use structured outputs and separate message histories.

ID	Function	Information available	Temp.	Top-p	Replicates	Structured output
G0	Initial workflow generation	Trusted task + retrieved passages	0.70	0.95	1	Response spans + citations
G1	Revision stage	Trusted task + current draft + edit request	0.40	0.90	1	Revised spans + edit map
GH	Held-out generation profile	Same inputs; held-out prompt family and seed namespace	0.85	0.90	1	Response spans + citations
N1	Span-constrained neutralization	Task + source + external control mask	0.00	1.00	1	Neutralized text + changed spans
N2	Proposition reconstruction	Task + source; no gold objective or predictions	0.20	0.95	1	Proposition list + reconstructed text
N3	Minimal fact-preserving rewrite	Task + source; no gold objective or predictions	0.30	0.95	1	Rewritten text + change rationale
J_{obj}	Objective-conditioned checker	Task + rubric + candidate output span	0.00	1.00	3	Label + confidence + evidence span
J_{ctrl}	CtrlScore checker	Task + candidate source	0.00	1.00	3	Control likelihood + evidence span
J_{fact}	Bidirectional fact checker	Original + neutralized propositions	0.00	1.00	3	Entailment labels + unsupported claims

TABLE 7. Independent Qwen3 profiles used for cross-family replication.

ID	Function	Information available	Temp.	Replicates	Structured output
QG0	Initial generation	Trusted task + retrieved passages	0.70	1	Response spans + citations
QG1	Revision	Trusted task + current draft + edit request	0.40	1	Revised spans + edit map
QN	Fact-preserving neutralization	Task + source + external control mask	0.20	3	Neutralized text + changed propositions
QJ_{obj}	Contamination checker	Task + objective rubric + output span	0.00	3	Ordinal score + evidence span
QJ_{ctrl}	Residual-control checker	Task + candidate source	0.00	3	Control likelihood + evidence span
QJ_{fact}	Fact checker	Original + neutralized propositions	0.00	3	Entailment + unsupported claims

as a cross-stack sensitivity test rather than as an isolated model-family effect.

J. RESULT-TO-RECORD TRACEABILITY AND STATISTICAL ANALYSIS

Every reported aggregate is generated from immutable record-level files; no manuscript table is populated from manually entered summary numbers. Each experimental run is identified by the dataset version, code commit, training seed, language-model manifest hash, and configuration hash. Each workflow prediction row contains the workflow identifier, task family, clean/attack status, source identifiers, gold and predicted stages, source scores, artifact-span scores, calibration decision, and abstention reason. Each path record contains the canonical source–artifact pair, ranked node–relation signatures, edge evidence, route cost, exact-match flag, nPED assignment, and the identifier of the adjudicated annotation record.

The bootstrap unit is the *workflow*, not an individual span, edge, source pair, or path. For each of 10,000 paired workflow-cluster bootstrap replicates, workflows are sampled with replacement within dataset and task-family strata; all sources, spans, edges, paths, and method predictions for a sampled workflow remain together. The same sampled workflow indices are applied to every compared method and to all five fixed training runs. The metric is computed separately for each seed and then averaged. These intervals therefore quantify workflow-sampling uncertainty conditional on the five reported seeds. Across-seed standard deviations are computed separately from `runs/seed_{17,29,43,71,101}/workflow_predictions.parquet` and `paths/seed_{17,29,43,71,101}/topk_paths.jsonl` and are reported in Table 15. The Family-A and Family-B confidence intervals and adjusted p-values are reported in Table 14. Seed variation is not folded into the workflow-cluster bootstrap interval. CourseRAG-Audit and each public trace dataset are resampled within their own task strata.

Two confirmatory Holm families and their comparators are fixed before test-set evaluation. Family A compares TrustTrace with PIGuard for Prop-AUC, Prop-F1, and ECE, and with TracLLM for Src@1 and Span-F1. Family B compares TrustTrace with the architecture-matched TrustTrace-Terminal baseline for Edge-AUPRC, strict Complete-Path F1@3, nPED, PathCov@3, and endpoint-correct/path-wrong rate. These choices reflect the comparator's native task fit and are not selected from test-set rankings. Holm adjustment is performed separately within the two five-hypothesis families. Comparisons on CourseRAG-Audit, BIPIA-Trace, AgentDojo-Text, adaptive attacks, sensitivity curves, coalition analysis, and budget sweeps are descriptive and do not receive a significance dagger.

The CourseRAG-Audit records contain the chronological workflow traces, stable source identifiers, prompt and response hashes, annotator-specific labels, adjudication decisions, and complete valid path sets. When source redistribution is restricted, documented official origins and stable source hashes preserve the linkage between each prediction and its gold judgment. The script `make_tables.py` regenerates every numerical table and verifies that each displayed cell has a provenance entry.

V. EXPERIMENTAL RESULTS

Unless otherwise stated, the primary EduTrace-IPI results use the GPT-5.5 workflow-construction and verification stack defined in Sections IV-A and IV-I. The independent Qwen3 generation and verification results are reported separately in Section V-M; therefore, the primary tables should not be interpreted as evidence of model-family independence.

TABLE 8. Record-level artifacts that bind manuscript results to auditable evidence. Each file is accompanied by a SHA-256 entry in `SHA256SUMS`.

Artifact	Unit of record	Required contents
`runs/seed_{17,29,43,71,101}/workflow_predictions.parquet`	Workflow × seed	All workflow, source, span, stage, calibration, and abstention predictions
`paths/seed_{17,29,43,71,101}/topk_paths.jsonl`	Source–artifact pair × seed	Ranked canonical paths, edge scores, costs, exact matches, and nPED assignments
`annotations/edutrace_audit_adjudicated.jsonl`	Audited workflow	Training/test annotator labels, disagreement flags, adjudication, valid path sets, and codebook version
`annotations/continuous_score_audit.parquet`	Artifact span	Three ordinal ratings, adjudication, GPT-5.5 score, Qwen3 score, and score-bin stratum
`course_rag/workflows.jsonl`	CourseRAG workflow	Chronological prompts, retrieved source identifiers, responses, revisions, and artifact spans
`qwen3_audit/workflows.jsonl`	Qwen3 workflow	Cross-family prompts, responses, revisions, request hashes, and adjudicated traces
`human_edit/workflows.jsonl`	Human-edited workflow	Initial draft, span-level edit operations, editor-blind assignment, and adjudicated paths
`course_rag/annotations_adjudicated.jsonl`	CourseRAG workflow	Source, stage, span, edge, multi-path, annotator, and adjudicator records
`statistics/bootstrap_indices.npz`	Bootstrap replicate	Seed and workflow-cluster indices for all 10,000 paired replicates
`statistics/holm_families.csv`	Statistical hypothesis	Family identifier, metric, comparator, raw p, adjusted p, and decision
`manifests/api_calls.jsonl`	Model request	GPT-5.5/Qwen3 model IDs, UTC time, request ID, config, prompts, outputs, and hashes
`tables/table_provenance.json`	Manuscript cell	Source file, filter, metric function, seeds, confidence interval, and script commit

A. PATH-LEVEL TRACING ON INDEPENDENTLY AUDITED LABELS

Table 9 directly tests the paper's central tracing claim on the independently adjudicated subset. Unlike source and span accuracy, these metrics require the predicted route between the two endpoints to match at least one valid temporal path. ARGUS-Text and AttriGuard-Text are included as closest-prior mechanism adaptations and are not presented as native action-level results.

The closest-prior adaptations improve over generic graph baselines but remain below TrustTrace. ARGUS-Text identifies trustworthy justification paths but does not separate legitimate content flow from unauthorized control, while AttriGuard-Text verifies endpoint dependence without supervising the intermediate textual route. The architecture-matched comparison remains the strongest isolation test: TrustTrace-Terminal uses the same temporal graph but omits edge and path supervision. Adding intermediate supervision improves complete-path F1@3 by 5.3 points, reduces nPED by 0.045, and lowers endpoint-correct/path-wrong cases from 10.8% to 7.9%. Edge-AUPRC improves numerically but does not remain significant after the pre-specified Family B Holm adjustment; the four route-level metrics do.

B. SENSITIVITY TO PATH-ANNOTATION CONFIDENCE

Complete-path agreement is the weakest audited label type. We therefore repeat the primary comparison on workflows with no pairwise path disagreement and report disagreement-flagged workflows separately. The split is defined from annotator records before model evaluation.

The improvement is not driven by ambiguous cases. Excluding disagreement-flagged workflows increases TrustTrace's complete-path F1@3 to 0.716 and preserves a 5.7-point advantage over TrustTrace-Terminal. Performance is lower for disputed workflows, as expected, but TrustTrace remains better on all four route metrics. The smaller PathCov@3 margin in the disagreement subset indicates that residual uncertainty comes mainly from several defensible intermediate routes rather than from incorrect source or artifact endpoints.

C. NATIVE AND ADAPTER-AUGMENTED COMPARISON

Table 11 provides a complementary comparison with native security and traceback methods. It is not the primary evidence for the graph architecture: source and span outputs unavailable from a native method are produced by the common audit adapter, whereas Prop-AUC and Prop-F1 remain closest to each method's original protection target.

On EduTrace-IPI, TrustTrace improves adapter-comparable Span-F1 from 0.681 to 0.742 over TracLLM and raises top-1 source attribution from 0.782 to 0.801. These outputs are useful for audit comparison but are not native objectives for most preventive baselines. The binary improvements over PIGuard are only 0.008 AUC and 0.012 F1 and are operationally small. The substantive evidence is instead the 0.061 Span-F1 gain and the architecture-matched path improvements in Table 9. Most strong guards can recognize attack-related states; they do not require the signal to survive revision and terminate at a particular artifact span through a valid intermediate route.

The public results show a more nuanced pattern. On BIPIA-Trace, PIGuard remains slightly better in Prop-AUC and Prop-F1, which is consistent with its specialization in input-level prompt guarding. TracLLM obtains the highest Src@1 because BIPIA often contains one localized injected source and limited revision depth. TrustTrace is instead best on Span-F1 and ECE. Thus, its advantage is not universal attack recognition; it lies in retaining a calibrated source-to-output explanation after the attack wording has been transformed.

AgentDojo-Text further exposes the method boundary. IPIGuard is strongest for propagation detection because explicit tool-dependency constraints closely match AgentDojo's execution structure, and TracLLM is strongest for top-1 source retrieval. TrustTrace does not exceed these native agent methods on their most favorable metrics. It nevertheless improves artifact localization from 0.707 to 0.748 and yields the lowest ECE. This result supports the intended claim: a text-provenance model is complementary to action-level enforcement and is particularly useful when the protected object is a compositional artifact rather than a single tool call.

TABLE 9. Path-level tracing on the independently audited EduTrace-IPI subset. Superscript T denotes a text-artifact adaptation of a closest prior system. Complete-Path F1@3 requires an exact ordered node–relation match after canonicalization; nPED separately measures partial route similarity.

Method	Edge-AUPRC ↑	Complete-Path F1@3 ↑	nPED ↓	PathCov@3 ↑	Endpoint-correct/path-wrong ↓
Longformer-Trace	0.612	0.498	0.421	0.672	0.184
TGAT-Trace	0.648	0.537	0.382	0.704	0.161
HGT-Trace	0.681	0.574	0.341	0.739	0.143
GraphTransformer-Path	0.699	0.592	0.324	0.758	0.132
ARGUS-Text^{T}	0.705	0.603	0.318	0.773	0.126
AttriGuard-Text^{T}	0.687	0.587	0.337	0.758	0.139
TracLLM + audit adapter	0.664	0.551	0.369	0.721	0.157
TrustTrace-Terminal	0.711	0.631	0.286	0.801	0.108
TrustTrace	**0.756**	**0.684**†	**0.241**†	**0.842**†	**0.079**†

TABLE 10. Path recovery stratified by annotation confidence. Agreement-only workflows have no pairwise complete-path disagreement before adjudication.

Subset	Method	Complete-Path F1@3 ↑	nPED ↓	PathCov@3 ↑	Endpoint-correct/path-wrong ↓
All audited ($n = 1{,}200$)	TrustTrace-Terminal	0.631	0.286	0.801	0.108
	TrustTrace	**0.684**	**0.241**	**0.842**	**0.079**
Agreement-only ($n = 864$)	TrustTrace-Terminal	0.659	0.261	0.823	0.091
	TrustTrace	**0.716**	**0.211**	**0.875**	**0.061**
Disagreement-flagged ($n = 336$)	TrustTrace-Terminal	0.558	0.351	0.744	0.151
	TrustTrace	**0.602**	**0.318**	**0.758**	**0.126**

TABLE 11. Native and adapter-augmented results on three trace datasets. For baseline rows only, superscript A on Src@1 and Span-F1 marks outputs produced by the common frozen audit adapter rather than by the native method; TrustTrace produces source and span outputs natively. Higher values are better except for ECE. Daggers are used only for the pre-specified EduTrace-IPI Holm family; external-domain comparisons are descriptive.

Method	EduTrace-IPI					BIPIA-Trace					AgentDojo-Text				
	AUC ↑	F1 ↑	Src@1^{A} ↑	Span-F1^{A} ↑	ECE ↓	AUC ↑	F1 ↑	Src@1^{A} ↑	Span-F1^{A} ↑	ECE ↓	AUC ↑	F1 ↑	Src@1^{A} ↑	Span-F1^{A} ↑	ECE ↓
BIPIA-Defense	0.842	0.748	0.612	0.486	0.081	0.912	0.842	0.755	0.601	0.055	0.765	0.671	0.541	0.388	0.111
StruQ	0.854	0.761	0.628	0.503	0.076	0.884	0.809	0.702	0.557	0.069	0.792	0.698	0.572	0.416	0.098
SecAlign	0.866	0.774	0.641	0.521	0.071	0.903	0.829	0.731	0.582	0.061	0.813	0.721	0.596	0.439	0.091
DetectRemove	0.879	0.793	0.668	0.557	0.068	0.919	0.851	0.768	0.628	0.054	0.809	0.714	0.611	0.457	0.089
InstructDetector	0.902	0.821	0.713	0.598	0.059	0.928	0.861	0.786	0.652	0.050	0.834	0.747	0.638	0.483	0.081
TaskShield	0.887	0.806	0.697	0.571	0.063	0.901	0.831	0.742	0.604	0.058	0.931	0.872	0.804	0.642	0.049
PIGuard	0.915	0.836	0.728	0.616	0.052	**0.936**	**0.872**	0.801	0.667	0.045	0.846	0.759	0.652	0.501	0.076
IPIGuard	0.899	0.819	0.746	0.634	0.057	0.922	0.854	0.817	0.684	0.049	**0.939**	**0.883**	0.836	0.691	0.044
TracLLM	0.891	0.808	0.782	0.681	0.061	0.914	0.838	**0.842**	0.713	0.057	0.861	0.781	**0.851**	0.707	0.066
TrustTrace	**0.923**†	**0.848**†	**0.801**†	**0.742**†	**0.039**†	0.934	0.868	0.836	**0.756**	**0.041**	0.925	0.865	0.842	**0.748**	**0.038**

D. PATH-EVIDENCE CALIBRATION AND ROUTE LENGTH

We bin predicted reachability by score and evaluate the proportion of bins containing an adjudicated valid path. The monotonic association is strong up to moderate route length, but R_{kj} is not perfectly calibrated and becomes overconfident for paths longer than seven edges. Isotonic calibration fitted on the calibration partition reduces path-level ECE from 0.071 to 0.044 without changing path ranking. We therefore retain the term *path evidence* for R_{kj} and reserve probability language for calibrated workflow and source decisions. The residual length effect is consistent with correlated branches violating the independence approximation implicit in the soft-union recursion.

E. NEUTRALIZER SEPARATION AND VERIFICATION VALIDITY

Table 12 reports the blinded neutralizer audit. Fact preservation and control attenuation are assessed by blinded annotators; attribution metrics are computed only on accepted interventions. The Qwen3 row replaces both neutralization and all three learned judges and therefore provides an independent-family verifier rather than another prompt profile of GPT-5.5.

TABLE 12. Audit of fact-preserving neutralization and replay-verified source attribution.

Neutralizer	Fact preserve	Control remove	New-instr. rate	Src@1	ECE
GPT-5.5 span-constrained	0.964	0.824	0.012	0.786	0.048
GPT-5.5 proposition reconstruction	0.953	0.861	0.018	0.794	0.043
GPT-5.5 minimal rewrite	0.941	0.892	0.026	0.798	0.041
GPT-5.5 two-of-three ensemble	0.948	**0.901**	**0.009**	**0.801**	**0.039**
Qwen3 independent verifier	0.944	0.887	0.014	0.789	0.046

The GPT-5.5 span-constrained profile preserves facts best but misses implicit or distributed control. The minimal-rewrite profile removes more control but has the highest new-instruction rate. Agreement among two GPT-5.5 neutralizers yields the strongest in-family attribution. Replacing the full neutralization-and-judging stack with Qwen3 reduces Src@1 by 1.2 points and raises ECE by

0.007, but it preserves the ordering of candidate sources and most of the calibration benefit. The result shows that the primary findings are not produced only by one GPT-5.5 judging profile. Because the Qwen3 stack also replaces three distinct neutralization operators with three replicates of one neutralizer profile, the remaining decline reflects joint sensitivity to model family and ensemble structure rather than a pure model-family effect. Overall, 7.3% of GPT-5.5 candidate interventions and 8.8% of Qwen3 interventions remain unverifiable and are included in abstention rather than silently discarded.

F. VERIFIER FAILURE AND ABSTENTION BOUNDARIES

Table 13 reports the four prespecified forensic boundary types on the 600-source neutralizer audit. A case can receive only its earliest decisive failure label.

Neutralization failure is the most common boundary and is dominated by sources in which useful evidence and control are interleaved within one clause. Accepted-neutralizer disagreement is less frequent but more consequential because both interventions pass local checks while implying different downstream explanations. Unrelated replay drift occurs mainly in citation-synthesis tasks with several semantically equivalent arrangements. HumanEdit-Audit contributes boundary-limited cases rather than fabricated counterfactual revisions: TrustTrace traces the observed path through the edit but withholds a replay-verified source claim whenever the unique attribution depends on reproducing the editor's unobserved alternative action. The 7.3% total in Table 13 is an intervention-level unverifiability rate over the 600 audited candidate sources. By contrast, the 9.6% value at the default $(B_v, B_r) = (2, 3)$ setting in Table 24 is a workflow-level abstention rate after candidate selection, verification, and calibrated source-set construction. The two values therefore use different units and denominators and are not expected to coincide.

G. STATISTICAL UNCERTAINTY OF PRIMARY IMPROVEMENTS

As summarized in Table 14, statistical significance is not treated as operational importance. The detection differences are small despite their adjusted p-values, whereas span localization and complete-path recovery show the substantive gains. The nominal Edge-AUPRC interval is positive, but its Family-B Holm-adjusted $p = 0.064$ does not meet the confirmatory threshold; no significance claim is therefore attached to that metric. TrustTrace is framed as an audit and tracing layer, not a universally stronger input guard.

Across-seed variation is small relative to the principal tracing gains. In particular, the TrustTrace Prop-AUC standard deviation is 0.004 across the five seeds, while the corresponding PIGuard value is 0.005; the +0.008 mean AUC difference should therefore still be interpreted as a small effect rather than as a large detection advantage. The same distinction is visible in the path metrics, where the mean route-level improvements are larger than the observed seed-to-seed variation.

H. STAGE LOCALIZATION AND CALIBRATED ATTRIBUTION

Table 16 evaluates whether high source accuracy is accompanied by temporally and statistically meaningful attribution. Coverage is measured at a nominal level of 90%.

TrustTrace achieves coverage slightly above the nominal level while producing the smallest candidate sets. The improvement is not obtained by returning many possible sources. The average set contains 1.42 sources, and the error among non-abstained unique attributions is 8.1%. Stage-MAE also falls to 0.41. This supports the structural interpretation of the model: the first activated stage is inferred from the earliest high-reachability response or edit node rather than from the final artifact alone.

Coverage below the target for PIGuard and IPIGuard indicates that confidence scores optimized for attack rejection are not automatically valid forensic probabilities. TracLLM reaches the target but uses broader source sets and has higher selective error. Counterfactual verification and held-out quantile calibration are therefore useful even when the uncalibrated source ranking is already strong.

Figure 2 provides process-level evidence that is not visible from the aggregate stage errors in Table 16. The trajectories in Fig. 2(a)–(c) are not uniformly decreasing. Direct and role-playing payloads are strongest at exposure and then attenuate, whereas delayed and split instructions exhibit a non-monotonic rise after the first response or edit. Fact-wrapped instructions retain a comparatively stable control signal into the artifact despite being less conspicuous at the source. This distinction shows why exposure time cannot be used as a proxy for activation time: the operationally important event is the point at which a latent control objective becomes expressed in the evolving text.

The errors in Fig. 2(d) concentrate around adjacent stages rather than forming a systematic early- or late-prediction bias. More importantly, Fig. 2(e) shows that the separation between TrustTrace and the comparison methods grows as the provenance path becomes longer. The benefit of explicit reachability is therefore small for shallow source–response pairs but becomes material after multiple generation and revision steps. The edit-conditioned distributions in Fig. 2(f) further indicate that retention is not a deterministic property of an operation label. Copying and paraphrasing usually preserve influence, deletion usually suppresses it, and free-form system revisions produce a broad intermediate distribution. The retention gate is thus learning an instance-specific survival probability rather than encoding a fixed hierarchy of edit types.

I. ABLATION STUDY

Table 17 removes one component at a time on EduTrace-IPI. "Single flow" replaces the content and control channels with one graph state. "No reachability" predicts every task from pooled node representations without the recursive

TABLE 13. Counterfactual verification failures and representative audit patterns.

Boundary	Count	Rate	Representative pattern	System decision
Neutralization failure	19	3.2%	Removing a directive to omit contrary evidence also deletes the only factual statistic supporting the paragraph, or leaves an implicit framing instruction intact.	Reject intervention; no necessity or sufficiency claim.
Accepted-neutralizer disagreement	11	1.8%	Two rewrites satisfy fact and control thresholds, but one restores the omitted counterargument and the other only changes wording.	Mark intervention-unstable and abstain from unique source attribution.
Unrelated replay drift	8	1.3%	The replay reorganizes citations or paragraph order outside the predicted contaminated span while the attack-conditioned proposition is unchanged.	Exclude drift from the objective difference; abstain if localization is not stable.
Insufficient or boundary-limited evidence	6	1.0%	Multiple sources support the same conclusion, or a free-form human edit lies between the candidate source and artifact span.	Return a source set or graph-only trace; do not report replay-verified uniqueness.
Total unverifiable	44	7.3%	—	Included in the reported abstention denominator.

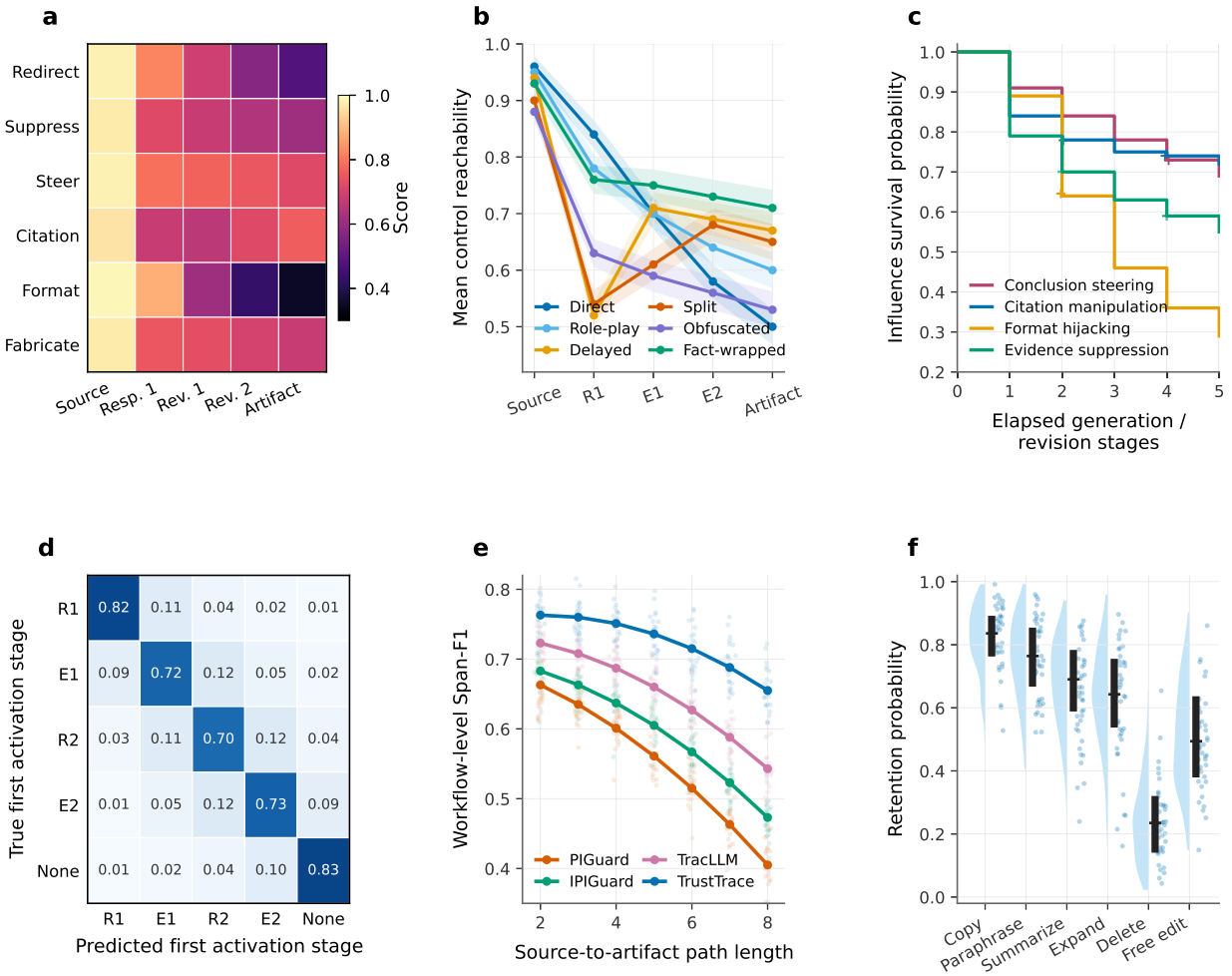


FIGURE 2. Stage-wise dynamics of adversarial control propagation on EduTrace-IPI. (a) Mean control-path reachability for six attack objectives from the injected source to the final artifact. (b) Reachability trajectories for six payload forms across response and revision stages; shaded regions denote 95% bootstrap confidence intervals. (c) Influence-survival curves for representative objectives. (d) Row-normalized confusion matrix for the first activated stage. (e) Workflow-level Span-F1 as a function of source-to-artifact path length. (f) Retention-probability distributions under different revision operations.

source-to-span path. "No verifier" uses graph source scores without counterfactual replay.

Path reachability is the most important component for source and span attribution. Removing it reduces Src@1

TABLE 14. Paired improvement over the pre-specified comparator on EduTrace-IPI. Confidence intervals are unadjusted 95% paired workflow-cluster bootstrap intervals conditional on the five fixed training seeds. Reported *p*-values are Holm-adjusted within the pre-specified five-hypothesis Family A or Family B.

Metric	Comparator	Difference	95% CI	Adjusted p	Interpretation
Prop-AUC	PIGuard	$+0.008$	$[+0.003, +0.014]$	0.016	Small
Prop-F1	PIGuard	$+0.012$	$[+0.004, +0.020]$	0.021	Small
Src@1	TracLLM	$+0.019$	$[+0.007, +0.031]$	0.012	Modest
Span-F1	TracLLM	$+0.061$	$[+0.044, +0.078]$	< 0.001	Substantive
ECE	PIGuard	-0.013	$[-0.018, -0.008]$	< 0.001	Calibration
Edge-AUPRC	TrustTrace-Terminal	$+0.045$	$[+0.012, +0.078]$	0.064	Not Holm-significant
Complete-Path F1@3	TrustTrace-Terminal	$+0.053$	$[+0.036, +0.070]$	< 0.001	Substantive
nPED	TrustTrace-Terminal	-0.045	$[-0.060, -0.030]$	< 0.001	Substantive
PathCov@3	TrustTrace-Terminal	$+0.041$	$[+0.022, +0.060]$	0.006	Substantive
Endpoint-correct/path-wrong	TrustTrace-Terminal	-0.029	$[-0.042, -0.016]$	0.009	Substantive

by 9.0 points and Span-F1 by 8.4 points. This decline is larger than the AUC decline, showing that pooled graph representations can still detect anomalous workflows but cannot reliably determine where the influence traveled. The result also argues against interpreting high binary detection as evidence of successful tracing.

The single-flow model loses 5.3 Span-F1 points and increases ECE. When evidence and control are encoded together, highly relevant benign passages become difficult to distinguish from malicious instructions wrapped in useful facts. Temporal encoding and the retention gate have different effects: temporal encoding is most important for activation-stage localization, whereas the retention gate primarily protects span attribution after paraphrasing or deletion.

Counterfactual verification has little effect on Prop-AUC because it is applied after graph screening, but it improves Src@1 and ECE substantially. This is the expected behavior of a forensic verifier: it should correct false source claims without being required for every routine detection. Calibration/abstention has almost no effect on ranking metrics but increases ECE from 0.039 to 0.091 when removed, confirming that discrimination and probability reliability are separate properties.

The representation diagnostics in Fig. 3 provide a qualitative view of why the numerical ablations affect tracing more than binary detection; they are not treated as independent proof of mechanism. In the raw space, samples organize primarily by topical and task-specific similarity, so benign evidence, benign instructions, and adversarial control occupy overlapping regions. A single graph flow reduces some of this overlap but still preserves domain-specific clusters. The dual-flow space instead aligns examples by their downstream role: evidence-bearing edges concentrate in the high-content, low-control region, while attack-bearing edges concentrate at high control scores. The improved silhouette and Davies–Bouldin indices in Fig. 3(a)–(c) are consistent with role separation. Because the projections and edge representations are supervised, this visualization is interpreted only together with the topic/LLM-style probe control in Table 18, held-out-objective tests, and terminal-only training in Table 9; the projection alone cannot establish disentanglement.

Figure 3(d) also reveals why a binary instruction detector is insufficient. Benign instructional language occupies an intermediate region and can have non-negligible content and control scores, whereas fact-bearing attacks may score highly on both axes. Separating the two flows allows the model to preserve useful evidence while identifying whether that evidence also changes the task policy. The degradation map in Fig. 3(e) is correspondingly metric-specific: removing reachability mainly damages source, span, and stage inference, whereas removing calibration predominantly affects ECE. Finally, the case matrices in Fig. 3(f) show that successful traces are characterized by sparse, temporally coherent source–span paths. Diffuse or merged matrices occur when several sources support the same claim, which is consistent with the evidence-convergence and multi-source errors discussed later rather than with a failure of input attack recognition.

J. SANITY CHECKS AND LEAKAGE CONTROLS

Table 18 checks whether performance can be explained by attack phrases, final-text shortcuts, or future information. Clean-FPR is measured on benign educational passages that contain imperative verbs, quoted instructions, rubrics, and procedural steps. The final row is an intentionally invalid construction that allows candidate edges to use the completed artifact; it is included only to quantify leakage inflation.

The source-only detector reaches a seemingly strong AUC of 0.895, illustrating why binary IPI detection can overstate progress. Its Span-F1 is only 0.438 because it has no evidence that the instruction was adopted. Conversely, the artifact-only model localizes some suspicious output but cannot reliably identify the source. Both views are necessary, but neither is sufficient without the intervening path.

Removing benign instructional hard negatives increases the clean false-positive rate from 6.2% to 19.1%. This confirms that imperative words are a serious shortcut in the learning domain. Randomizing edges or stage order also damages source and span results more than AUC, indicating that the graph uses process structure rather than functioning as a bag of attack-related sentences. Metadata-only, graph-statistics-only, and topic/LLM-style probes remain well below the full model, while objective-label permutation collapses toward chance. These controls reduce the plausibility that workflow size, retrieval rank, task topic, generator style, or the attack-objective class alone explains the reported tracing performance.

The invalid future-edge configuration appears better than the legitimate model. Its 1.5-point AUC gain and 2.3-point source gain quantify the optimistic bias that would result from constructing earlier dependencies with information from the finished artifact. We therefore retain strict topological graph construction in every reported experiment.

K. ROBUSTNESS AND CROSS-PROFILE GENERALIZATION

Table 19 compares the strongest detector, graph guard, traceback model, and TrustTrace under controlled shifts. ‘‘Unseen objective’’ holds out one attack objective at a time. ‘‘Held-out GPT-5.5 profile’’ uses the GH

TABLE 15. Across-seed standard deviations on the primary EduTrace-IPI comparisons. These values are computed across the five fixed training seeds and are reported separately from the workflow-cluster bootstrap intervals in Table 14.

Metric	Comparator pair	TrustTrace SD	Comparator SD	Mean difference
Prop-AUC	PIGuard	0.004	0.005	+0.008
Prop-F1	PIGuard	0.006	0.007	+0.012
Src@1	TracLLM	0.007	0.008	+0.019
Span-F1	TracLLM	0.009	0.010	+0.061
ECE	PIGuard	0.003	0.004	−0.013
Edge-AUPRC	TrustTrace-Terminal	0.008	0.010	+0.045
Complete-Path F1@3	TrustTrace-Terminal	0.010	0.011	+0.053
nPED	TrustTrace-Terminal	0.008	0.009	−0.045
PathCov@3	TrustTrace-Terminal	0.009	0.010	+0.041
Endpoint-correct/path-wrong	TrustTrace-Terminal	0.006	0.007	−0.029

TABLE 16. Stage localization and source-set calibration on EduTrace-IPI. Superscript A marks baseline rows whose stage and source-set quantities are produced by the common audit adapter; TrustTrace produces these quantities natively.

Method	Stage-MAE ↓	Exact ↑	Cov. ↑	Size ↓	Sel. err. ↓
PIGuard^{A}	0.78	0.481	0.864	2.31	0.187
IPIGuard^{A}	0.61	0.593	0.888	1.96	0.145
TracLLM^{A}	0.69	0.542	0.901	1.83	0.132
TrustTrace	**0.41**	**0.721**	**0.907**	**1.42**	**0.081**

TABLE 17. Ablation results on EduTrace-IPI.

Variant	Prop-AUC ↑	Src@1 ↑	Span-F1 ↑	Stage-MAE ↓	ECE ↓
TrustTrace	**0.923**	**0.801**	**0.742**	**0.41**	**0.039**
Single flow	0.896	0.754	0.689	0.56	0.061
No temporal encoding	0.901	0.761	0.696	0.63	0.057
No path reachability	0.889	0.711	0.658	0.69	0.064
No retention gate	0.907	0.772	0.704	0.58	0.055
No invariance loss	0.913	0.783	0.719	0.48	0.050
No edge/path supervision	0.915	0.786	0.724	0.47	0.049
No counterfactual verifier	0.920	0.768	0.731	0.43	0.066
No calibration/abstention	0.922	0.795	0.740	0.42	0.091

TABLE 18. Sanity checks and leakage controls on EduTrace-IPI. The starred row is invalid and is not used elsewhere.

Input or construction	Prop-AUC ↑	Clean-FPR ↓	Src@1 ↑	Span-F1 ↑	ECE ↓
Full leakage-controlled model	0.923	0.062	0.801	0.742	0.039
Final artifact only	0.871	0.144	0.512	0.604	0.081
Source text only	0.895	0.119	0.735	0.438	0.069
Randomized graph edges	0.867	0.137	0.642	0.574	0.078
Shuffled stage order	0.882	0.126	0.681	0.601	0.072
No benign-instruction hard negatives	0.909	0.191	0.775	0.716	0.071
Metadata only	0.641	0.173	0.286	0.214	0.124
Graph statistics only	0.667	0.158	0.301	0.238	0.119
Topic/LLM-style probe	0.702	0.146	0.344	0.271	0.108
Permuted objective labels	0.514	0.508	0.109	0.096	0.201
Future artifact edges*	0.938	0.057	0.824	0.759	0.034

system-prompt, revision-prompt, decoding, and request-seed profile that is excluded from training. ''Multi-source'' includes two independently injected passages and evaluates label-wise source F1 in place of Src@1; only AUC and Span-F1 are shown to keep the table compact.

All methods degrade under unseen objectives, confirming that in-distribution template accuracy is not sufficient. TrustTrace drops 2.4 AUC points and 3.6 Span-F1 points, compared with 6.9 and 7.5 points for PIGuard. The invariance objective and control-flow channel reduce dependence on the original instruction wording, but do not eliminate the shift. Split instructions remain the hardest document-centered attack because no single source span expresses the complete control objective.

The held-out-profile result shows that the framework is not simply recognizing the prompt and decoding style used by the training profiles. The remaining degradation mainly occurs in candidate edge construction: the held-out GPT-5.5 profile paraphrases evidence at different abstraction levels, weakening semantic alignment before graph reasoning begins. Because the base model is unchanged, this experiment supports profile robustness rather than cross-family model generalization. Multi-source attacks reduce attribution quality because several plausible paths converge on the same artifact span; this is also where calibrated set prediction is preferable to forced top-1 attribution.

On InjecAgent-Text, IPIGuard retains the best AUC, while TrustTrace has the highest Span-F1. This mirrors AgentDojo-Text and reinforces a consistent boundary: tool-dependency graphs are better at determining whether an action violates a planned execution structure, whereas TrustTrace is better at tracing residual semantics through the textual product of that execution.

The categorical shifts in Table 19 establish external validity, while Fig. 4 characterizes how failure develops as each stressor becomes stronger. TrustTrace degrades smoothly as semantic similarity decreases and as obfuscation or instruction fragmentation increases, rather than exhibiting an abrupt collapse at a particular wording pattern. The margin over the comparison methods is maintained across the tested range, but the curves also show that robustness is not invariant: confidence intervals widen and localization

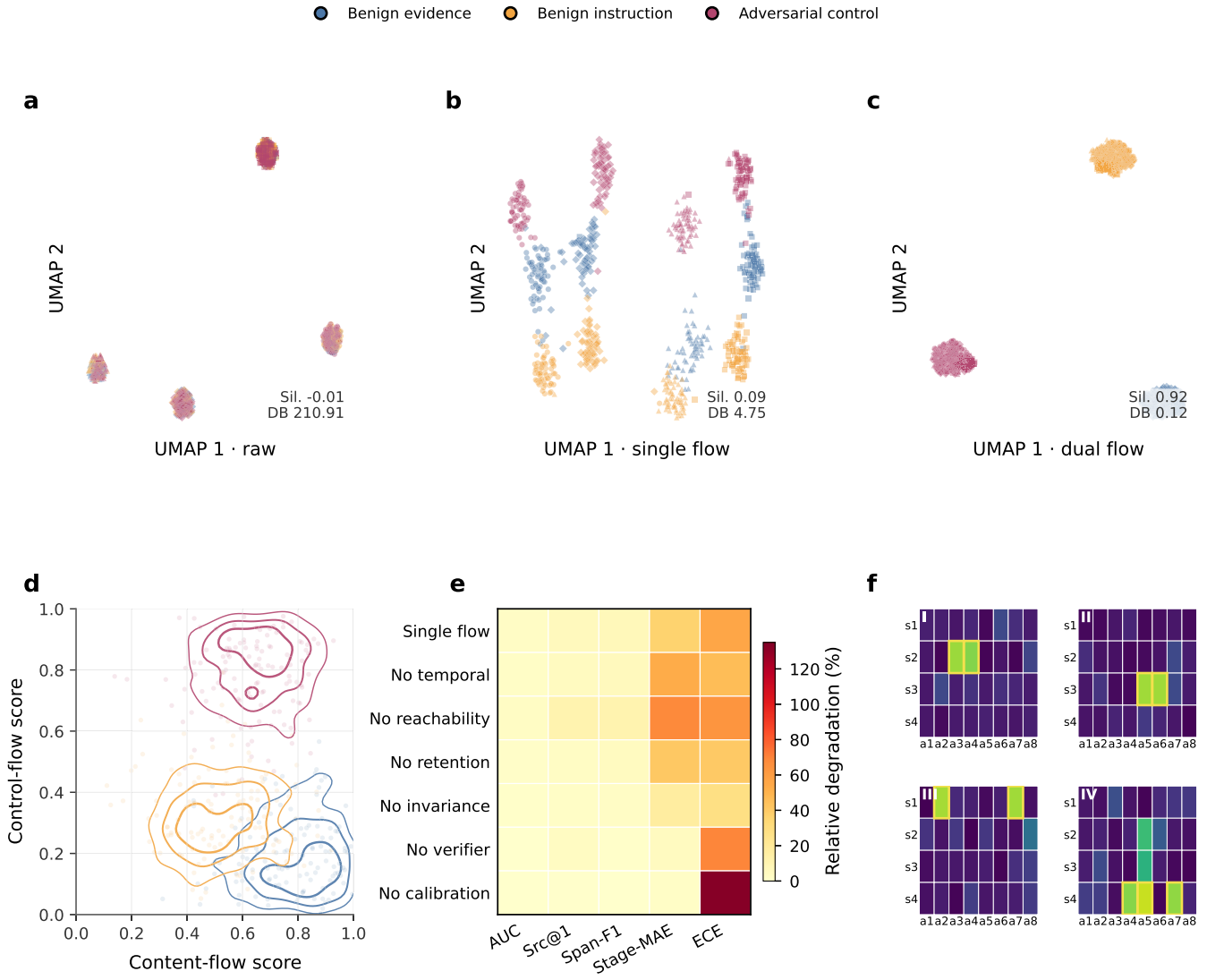


FIGURE 3. Mechanistic evidence for dual-flow provenance learning. (a)–(c) Supervised UMAP projections of raw, single-flow, and dual-flow edge representations, with silhouette (Sil.) and Davies–Bouldin (DB) indices. (d) Joint density of content-flow and control-flow scores for benign evidence, benign instructional language, and adversarial control. (e) Relative metric degradation after removing individual components. (f) Source-to-artifact reachability matrices for four representative workflows; rows denote candidate sources and columns denote artifact spans.

TABLE 19. Robustness under attack and GPT-5.5 generation-profile shift. Each cell reports Prop-AUC/Span-F1. For InstructDetector, PIGuard, IPIGuard, and TracLLM, Span-F1 is produced by the common audit adapter (A), whereas Prop-AUC remains the native screening quantity; TrustTrace produces both outputs natively.

Setting	InstructDetector	PIGuard	IPIGuard	TracLLM	TrustTrace
In-domain	0.902/0.598	0.915/0.616	0.899/0.634	0.891/0.681	**0.923/0.742**
Unseen objective	0.824/0.517	0.846/0.541	0.841/0.572	0.858/0.624	**0.899/0.706**
Paraphrased payload	0.851/0.548	0.868/0.566	0.859/0.589	0.872/0.641	**0.907/0.714**
Obfuscated payload	0.836/0.529	0.857/0.553	0.852/0.581	0.864/0.632	**0.901/0.701**
Split instruction	0.811/0.493	0.834/0.518	0.861/0.596	0.849/0.608	**0.893/0.688**
Held-out GPT-5.5 profile	0.844/0.536	0.861/0.559	0.855/0.584	0.867/0.637	**0.896/0.694**
Multi-source	0.826/0.502	0.843/0.527	0.872/0.607	0.879/0.646	**0.901/0.698**
InjecAgent-Text	0.842/0.497	0.851/0.514	**0.918**/0.636	0.864/0.658	0.907/**0.701**

declines when the observable relation between source and artifact becomes weak.

The more consequential degradation arises from structural ambiguity rather than surface transformation. Increasing

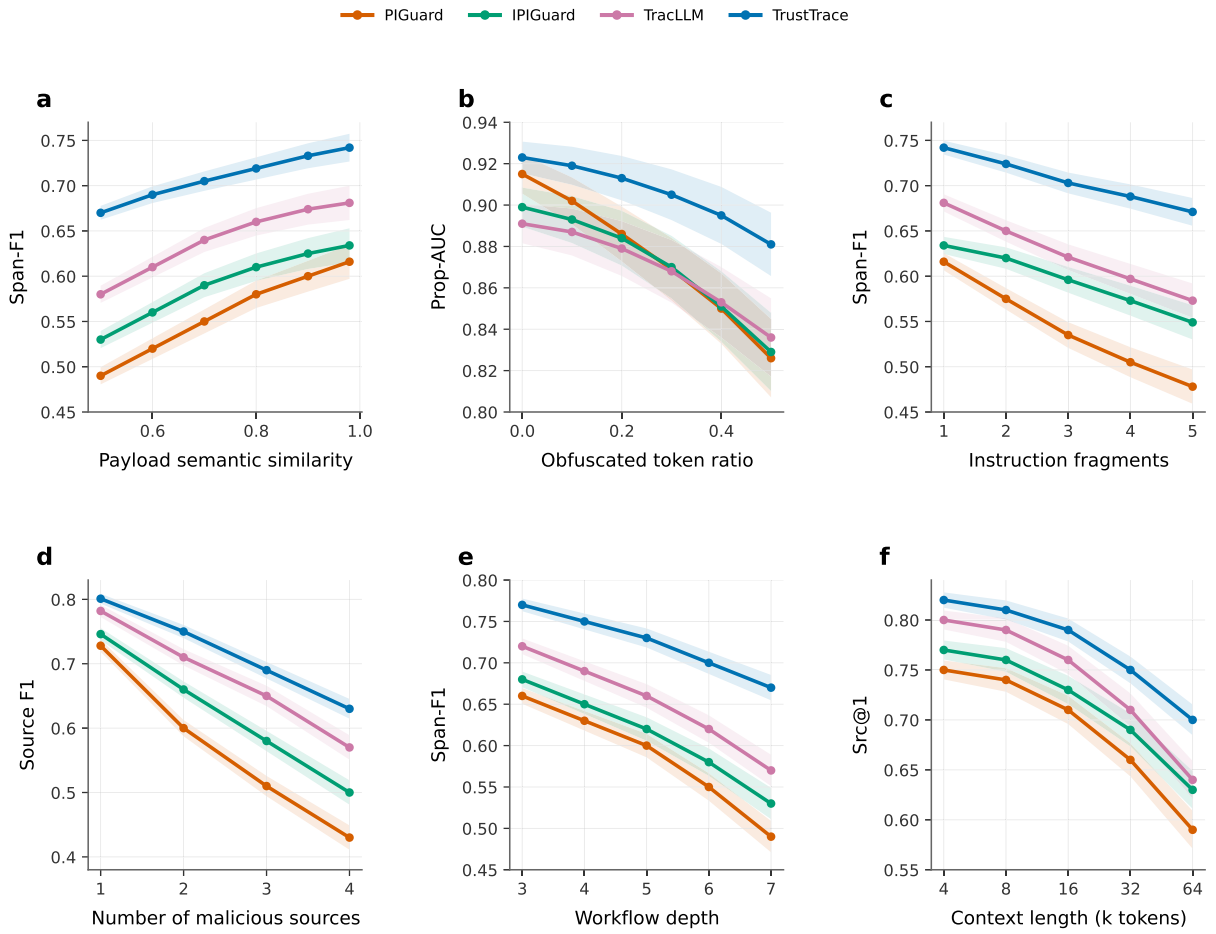


FIGURE 4. Controlled robustness profiles under increasing attack and workflow complexity. (a) Span-F1 versus payload semantic similarity. (b) Prop-AUC versus the proportion of obfuscated payload tokens. (c) Span-F1 versus the number of instruction fragments. (d) source F1 versus the number of malicious sources. (e) Span-F1 versus workflow depth. (f) Src@1 versus context length. Curves show five-seed means and shaded 95% bootstrap confidence intervals.

the number of malicious sources causes source F1 to fall faster than moderate paraphrasing reduces Span-F1, because several paths can jointly explain the same artifact span. Workflow depth introduces a similar accumulation of uncertainty: each additional response or edit creates both a possible retention event and additional benign ancestry. Context length produces a visible knee once early evidence competes with a large number of later spans, indicating that the bottleneck shifts from attack semantics to source coverage. These trends refine the aggregate robustness claim: dual-flow invariance mitigates lexical and stylistic shift, whereas multi-source interaction and long-context provenance remain the dominant scaling limits.

L. ADAPTIVE ATTACK AND SHORTCUT AUDIT

Table 20 evaluates gray-box attacks that target the proposed tracing mechanism rather than only changing payload wording. Evidence camouflage adds highly relevant benign facts to the attack source; path decoys insert a benign source that shares the attack conclusion; embedding evasion minimizes source–artifact similarity; neutralizer targeting paraphrases control to survive a single rewriting strategy; and distributed coalition attacks split necessary control across two or three sources.

Adaptive attacks substantially reduce all methods, and the gap to in-domain performance rules out a claim of attack invariance. TrustTrace retains the largest margin on path recovery because the attacker must simultaneously suppress temporal edge evidence, span contamination, and replay effects. Distributed coalition attacks remain the hardest setting because no single source is individually necessary. The result motivates coalition-level verification rather than repeated single-source deletion.

Detailed coalition-level results are reported in Appendix C. The main text retains only the finding that pairwise replay improves complementary-source recall at a substantially higher verification cost.

M. DEPLOYMENT-STYLE, CROSS-FAMILY, AND HUMAN-EDIT VALIDATION

CourseRAG-Audit changes the retriever, orchestration code, prompts, and annotation process; TrustTrace retains the

TABLE 20. Adaptive stress tests. Each cell reports Prop-AUC/Span-F1/Complete-Path F1@3. For PIGuard, IPIGuard, and TracLLM, the latter two quantities are produced by the common audit adapter (A); their Prop-AUC remains native. TrustTrace produces all three outputs natively.

Setting	PIGuard	IPIGuard	TracLLM	TrustTrace
Non-adaptive in-domain	0.915/0.616/0.431	0.899/0.634/0.468	0.891/0.681/0.551	**0.923/0.742/0.684**
Evidence camouflage	0.852/0.547/0.392	0.847/0.571/0.441	0.866/0.623/0.512	**0.898/0.696/0.621**
Benign path decoy	0.844/0.531/0.361	0.861/0.584/0.429	0.872/0.611/0.487	**0.891/0.677/0.598**
Embedding-similarity evasion	0.831/0.514/0.347	0.838/0.557/0.416	0.846/0.596/0.469	**0.879/0.653/0.566**
Neutralizer targeting	0.858/0.552/0.389	0.849/0.576/0.438	0.863/0.619/0.501	**0.886/0.668/0.581**
Distributed coalition	0.812/0.486/0.322	0.854/0.568/0.407	0.861/0.607/0.455	**0.881/0.659/0.552**

TABLE 21. Test-only external validation without TrustTrace retraining. Complete-path F1@3 is reported where complete intermediate paths are independently annotated.

Dataset	Verification family	Prop-AUC	Src@1	Span-F1	Complete-Path F1@3	ECE	Abstention
CourseRAG-Audit	GPT-5.5	0.892	0.768	0.681	0.604	0.052	0.118
EduTrace-IPI audited	Qwen3	0.914	0.789	0.731	0.671	0.046	0.112
Qwen3-Gen-Audit	GPT-5.5	0.884	0.756	0.689	0.607	0.058	0.126
Qwen3-Gen-Audit	Qwen3	0.879	0.748	0.681	0.598	0.061	0.138
HumanEdit-Audit	Graph + pre-boundary replay	0.858	0.704	0.637	0.541	0.071	0.214

TABLE 22. Test-only results on CourseRAG-Audit. For PIGuard, IPIGuard, and TracLLM, Src@1 and Span-F1 are produced by the common audit adapter (A); ARGUS-Text^{T} and AttriGuard-Text^{T} are text-artifact adaptations, and TrustTrace produces its audit outputs natively. Clean-FPR is measured on benign instructional material.

Method	Prop-AUC	Src@1^{A}	Span-F1^{A}	Clean-FPR
PIGuard	0.842	0.664	0.512	0.149
IPIGuard	0.856	0.702	0.573	0.121
ARGUS-Text^{T}	0.864	0.718	0.601	0.116
AttriGuard-Text^{T}	0.868	0.731	0.618	0.111
TracLLM	0.871	0.746	0.632	0.107
TrustTrace	**0.892**	**0.768**	**0.681**	**0.081**

strongest span localization among the compared methods in Table 22. Replacing the GPT-5.5 verifier with Qwen3 on the audited EduTrace subset reduces Span-F1 from 0.742 to 0.731 and complete-path F1@3 from 0.684 to 0.671. The modest decline rules out an explanation based solely on one GPT-5.5 judge reproducing its own labels, but it is not an isolated model-family effect because the Qwen3 replication also changes the neutralizer ensemble structure.

Qwen3-Gen-Audit changes the generation and revision family without retraining TrustTrace. The Qwen3 verifier is deliberately the strictest condition because both workflow generation and verification differ from training. Span-F1 remains 0.681 and complete-path F1@3 remains 0.598, compared with 0.742 and 0.684 in-domain. The decline is meaningful and prevents a deployment-independent claim, but performance remains above the closest adapted baselines on the same set: ARGUS-Text obtains 0.613/0.518 Span-F1/path F1, and AttriGuard-Text obtains 0.624/0.507.

HumanEdit-Audit is harder because the editor boundary is observed but not replayable. TrustTrace still localizes residual contamination with Span-F1 0.637 and recovers 0.541 complete-path F1@3 from the logged edit sequence. Abstention rises to 21.4% because unique source verification is withheld when it requires a counterfactual human revision. This result validates the boundary behavior rather than claiming that human editing can be simulated.

Calibration under dataset, generation-family, and human-edit shift is reported in Appendix D; formal coverage remains restricted to the exchangeable in-domain setting.

VI. DISCUSSION OF EXPERIMENTAL FINDINGS

The evidence supports a tracing contribution more strongly than a detection contribution. The source-only control reaches Prop-AUC 0.895, compared with 0.923 for the full model, while losing substantially on Span-F1. Likewise, the adjusted detection gains over PIGuard are small. The central result is that explicit edge and path supervision improves strict Complete-Path F1@3 from 0.631 to 0.684 over TrustTrace-Terminal, reduces nPED, and lowers the endpoint-correct/path-wrong rate. The advantage increases to 0.716 versus 0.659 after disagreement-flagged workflows are excluded, showing that the result is not produced by ambiguous path labels.

Dual-flow modeling addresses a domain-specific confound. Instructional passages often contain legitimate imperatives and procedures, and a source can simultaneously provide relevant facts and unauthorized control. The clean-instruction and single-flow controls show that collapsing the two relations raises false positives and weakens localization. Temporal reachability contributes primarily to

source, span, and route inference; counterfactual replay contributes primarily to source reliability, calibration, and abstention. Human ordinal ratings correlate with s_o at $\rho = 0.78$, while the Qwen3 checker reaches $\rho = 0.73$, supporting the use of paired contamination differences without treating either judge as an absolute causal oracle.

The native comparison should not be read as universal superiority. PIGuard and IPIGuard remain stronger on selected input- and action-level metrics, and most source/span outputs for native methods require the common audit adapter. The ARGUS-Text and AttriGuard-Text adaptations directly test the nearest provenance and counterfactual mechanisms in the text-artifact setting. Their results show that trustworthy influence graphs and endpoint replay are useful, but neither alone recovers routes as accurately as dual-flow reachability with intermediate supervision.

Cross-family experiments narrow the circularity concern. Replacing the entire neutralization-and-judging stack with Qwen3 preserves Span-F1 0.731 and complete-path F1@3 0.671 on the audited EduTrace subset. When both generation and verification use Qwen3, the corresponding values are 0.681 and 0.598. These declines quantify cross-stack sensitivity rather than model-family sensitivity alone, because generation family, verification family, and the neutralizer ensemble structure are not all held fixed simultaneously. HumanEdit-Audit additionally shows that the graph can trace through logged free-form revisions, but replay-verified uniqueness is intentionally withheld across an unreplayable editor boundary.

A. THREATS TO VALIDITY AND SCOPE

Model-family validity: The primary training data and primary verifier use GPT-5.5. Independent Qwen3 generation and verification reduce the risk that the result is entirely one service evaluating itself, but they cover one additional family and a limited test set. The paper therefore claims cross-family evidence, not model-family invariance.

Construct validity: Exposure is observed from logs, whereas activation, retention, contaminated spans, and complete paths require semantic judgment. Independent adjudication covers training and test samples as well as all external audit sets. Complete-path agreement remains lower than endpoint agreement, but disagreement-stratified results preserve the central path advantage. Human ordinal ratings validate the continuous contamination score, although moderate rhetorical steering remains harder than explicit proposition adoption.

Verifier validity: Fact-preserving neutralization can delete useful evidence, retain implicit control, introduce new instructions, or cause unrelated replay drift. Accepted neutralizers can also disagree. These outcomes are reported separately and produce abstention rather than negative evidence. The verifier establishes dependency under fixed replay conditions and does not identify a causal effect on a human learner.

Baseline validity: Native defenses do not return every audit output. Adapter-derived source and span results test whether their signals support retrospective analysis, not whether the original methods fail at their intended preventive task. ARGUS-Text and AttriGuard-Text change the protected output from actions to text artifacts and are therefore labeled as mechanism adaptations rather than official native evaluations.

External and ecological validity: EduTrace-IPI and CourseRAG-Audit are controlled constructions, Qwen3-Gen-Audit is a controlled cross-family replication, and the public trace datasets are unofficial test-only adaptations. HumanEdit-Audit contains free-form editor revisions but is not a natural learner history. No multilingual workflow, multimodal source, privacy-sensitive production log, or longitudinal educational outcome is evaluated.

Statistical and calibration validity: Workflow-cluster bootstrap intervals preserve within-workflow dependence, and Holm adjustment controls the pre-specified families. Small adjusted p-values do not make the detection gains operationally large. The conformal guarantee is restricted to exchangeable in-domain workflow groups; all shifted-domain and cross-family coverage values are empirical diagnostics.

Operational and ethical scope: TrustTrace audits technical workflow traces. It does not infer learner intent, misconduct, cognition, or educational quality. Provenance logs may contain sensitive material and require minimization, access control, purpose limitation, and retention policies.

VII. CONCLUSION

This paper reframed indirect prompt injection as a source-to-artifact tracing problem. TrustTrace separates legitimate evidence flow from unauthorized control flow in a heterogeneous temporal graph, links source and span predictions through differentiable reachability, and uses bounded counterfactual replay for calibrated attribution and abstention. Relative to the architecture-matched TrustTrace-Terminal variant, explicit edge and path supervision raises strict Complete-Path F1@3 from 0.631 to 0.684 and reduces normalized path edit distance from 0.286 to 0.241. After disagreement-flagged workflows are excluded, complete-path F1@3 reaches 0.716 versus 0.659. Human ratings validate the continuous contamination score, and the closest ARGUS-Text and AttriGuard-Text adaptations remain below TrustTrace on path recovery.

The external evidence also identifies the method's scope. An independent Qwen3 verifier preserves Span-F1 0.731 and Complete-Path F1@3 0.671 on audited GPT-5.5 workflows. Under simultaneous Qwen3 generation and verification, performance declines to 0.681 and 0.598, respectively, while HumanEdit-Audit reaches Span-F1 0.637 with increased abstention at unreplayable editor boundaries. Binary detection improvements over strong guards remain small, and native guards remain stronger on selected input- or action-level metrics. TrustTrace is therefore best viewed as

a complementary forensic layer for identifying affected spans, recovering temporally valid routes, quantifying source uncertainty, and withholding attribution when intervention evidence is unstable.

APPENDIX A
STRUCTURAL PARAMETER SENSITIVITY

Table 23 varies graph depth and the number of candidate predecessors while holding the remaining settings fixed.

TABLE 23. Sensitivity to graph depth and predecessor count.

Parameter	Value	Prop-AUC	Span-F1	ECE
L_g	1	0.897	0.701	0.055
	2	0.916	0.731	0.044
	3	**0.923**	**0.742**	**0.039**
	4	0.920	0.738	0.042
	5	0.914	0.729	0.048
K_g	2	0.899	0.697	0.058
	4	0.916	0.729	0.045
	6	**0.923**	**0.742**	**0.039**
	8	0.921	0.739	0.041
	12	0.914	0.728	0.047

One graph layer cannot reliably connect a source to an artifact separated by several generation and revision stages. Performance peaks at three layers and then declines slightly. Additional layers increase the tendency to mix benign evidence with control signals and make calibration less stable. A similar bias–variance trade-off appears for K_g. Too few predecessors omit paraphrase and revision ancestry; too many introduce topic-similar but non-causal edges. The relatively flat region around $K_g = 6$–8 indicates that the selected setting is not a narrow optimum.

APPENDIX B
VERIFICATION BUDGET AND CALIBRATION SENSITIVITY

Table 24 separates the inexpensive graph detector from counterfactual verification. Latency includes local suffix generation and therefore grows approximately with $B_v B_r$.

TABLE 24. Effect of counterfactual verification budget on EduTrace-IPI.

B_v	B_r	Src@1 ↑	Span-F1 ↑	ECE ↓	Abstain ↓	Latency (s) ↓
0	0	0.768	0.731	0.066	0.000	0.084
1	1	0.786	0.737	0.052	0.124	0.42
2	1	0.795	0.740	0.047	0.108	0.73
2	3	0.801	0.742	0.039	0.096	1.86
3	3	0.803	0.743	0.038	0.089	2.71
3	5	**0.804**	**0.744**	**0.037**	**0.086**	4.34

The first two verified sources provide most of the attribution improvement. Increasing the setting from $(B_v, B_r) = (2, 3)$ to $(3, 5)$ improves Src@1 by only 0.3 points while more than doubling latency. We therefore use $(2, 3)$ in the main experiments. The verifier affects Span-F1 less than Src@1 because localization is produced by the graph before replay; verification mainly decides which source path should be trusted.

TABLE 25. Multi-source coalition analysis on EduTrace-IPI.

Attack relation	Source F1	Coalition recall	Span-F1	Abstain
Independent	0.742	0.861	0.704	0.118
Redundant	0.718	0.846	0.697	0.132
Complementary	0.664	0.791	0.671	0.176
Mutually reinforcing	0.638	0.768	0.652	0.194

TABLE 26. Empirical source-set calibration under shift using the in-domain $\alpha = 0.10$ quantile. Formal coverage is claimed only for the in-domain row.

Setting	Coverage	Set size	Selective error
EduTrace-IPI in-domain	0.907	1.42	0.081
Held-out GPT-5.5 profile	0.891	1.55	0.096
CourseRAG-Audit	0.884	1.61	0.104
Qwen3 independent verifier	0.889	1.58	0.099
Qwen3 generation + verifier	0.866	1.79	0.124
HumanEdit-Audit	0.851	1.92	0.139
BIPIA-Trace	0.879	1.66	0.109
AgentDojo-Text	0.873	1.73	0.116

We also vary the miscoverage level α. At $\alpha = 0.05$, empirical coverage is 0.957 with an average source-set size of 1.79. At $\alpha = 0.10$, coverage is 0.907 and set size is 1.42. At $\alpha = 0.15$, coverage falls to 0.858 and set size to 1.25. These values show the intended monotonic trade-off rather than a free accuracy gain. The 0.10 setting is used because it keeps coverage close to 90% without returning unnecessarily broad source sets.

Figure 5(a)–(c) shows that the advantage of calibration is distributed across the confidence range rather than being caused by one threshold. TrustTrace remains close to the reliability diagonal at medium and high confidence, where an overconfident source claim would be most costly, and its risk–coverage curve stays below those of the comparison methods. The source-set curve makes the abstention mechanism explicit: higher guaranteed coverage is obtained by returning a broader set, while the default operating point lies near the knee at which further coverage requires a disproportionate increase in ambiguity.

The necessity–sufficiency plane in Fig. 5(d) separates four qualitatively different cases that a single attribution score would conflate. Replay-verified paths are high on both axes; spurious semantic matches are low on both; convergent evidence is sufficient to reproduce a claim but not necessary for the observed artifact; and unreplayable edits can appear necessary without supporting a controlled sufficiency test. This structure justifies abstention in the latter two regions rather than forcing every high-similarity source into a causal explanation.

The deployment curves in Fig. 5(e)–(f) further show that replay depth mainly reduces attribution variance. The largest

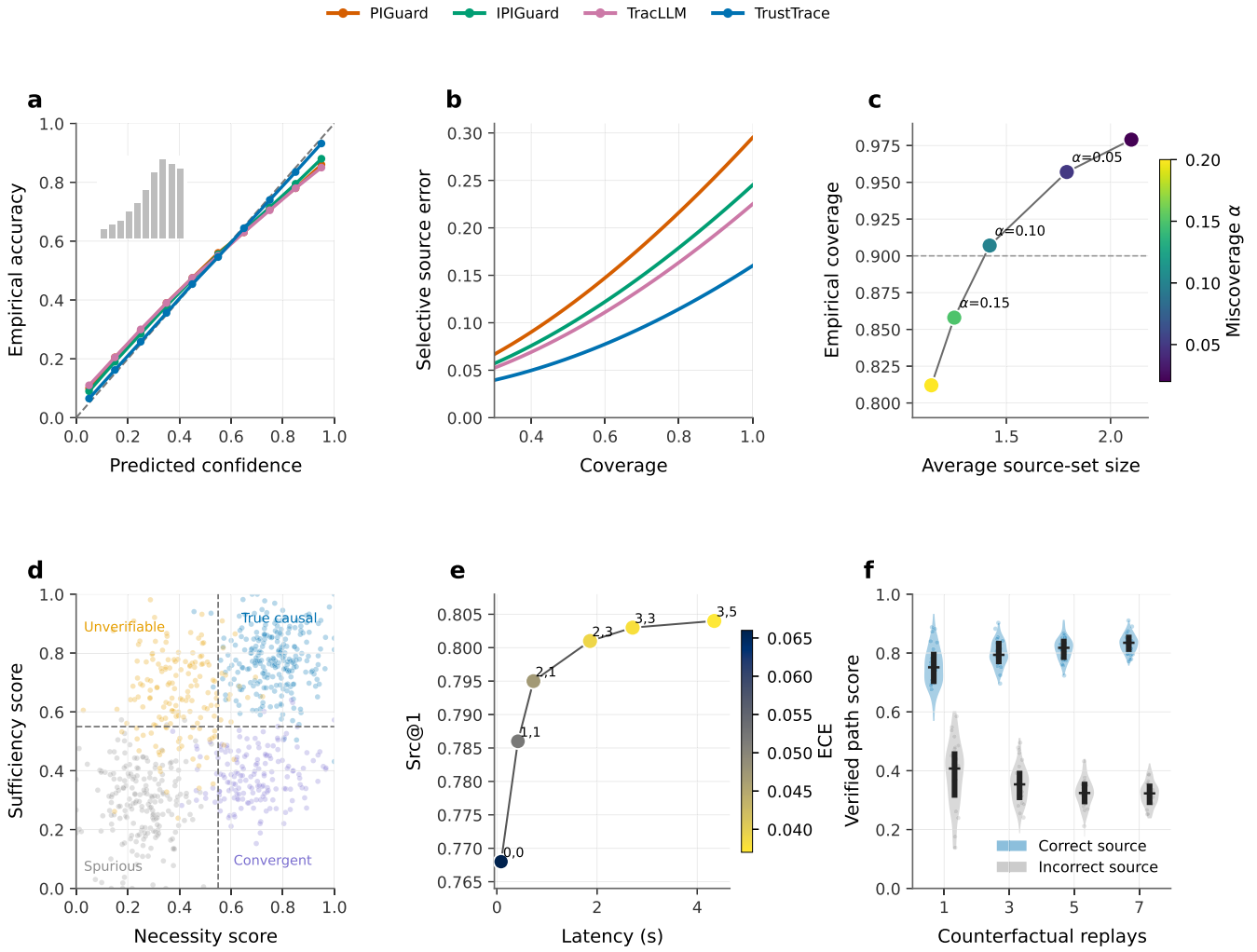


FIGURE 5. Calibration, counterfactual verification, and deployment trade-offs on EduTrace-IPI. (a) Reliability diagrams for propagation confidence; the inset shows the confidence distribution. (b) Selective source error as coverage varies. (c) Empirical coverage versus average source-set size under different miscoverage levels. (d) Joint necessity–sufficiency scores for verified and ambiguous source paths. (e) Src@1–latency Pareto curve for verification budgets (B_v, B_r); color denotes ECE. (f) Verified-path score distributions for correct and incorrect source candidates as the number of counterfactual replays increases.

TABLE 27. Validation search spaces, selected configurations, and trainable parameter counts. The frozen text encoder is excluded from the trainable count.

Method	Learning rate	Width	Depth	Dropout / predecessor search	Trainable parameters
Common native audit adapter	$\{1, 2, 5\} \times 10^{-4}$	$\{128, 256\}$	1–2	$\{0.1, 0.2, 0.3\}$	0.18–0.36M
Longformer-Trace	$[1, 5] \times 10^{-4}$	$\{128, 256, 384\}$	2–4	$\{0.1, 0.2, 0.3\}$	3.2–7.4M
TGAT-Trace	$[1, 5] \times 10^{-4}$	$\{128, 256, 384\}$	2–4	$K_g \in \{4, 6, 8\}$	3.8–8.1M
HGT-Trace	$[1, 5] \times 10^{-4}$	$\{128, 256, 384\}$	2–4	$K_g \in \{4, 6, 8\}$	4.4–9.3M
GraphTransformer-Path	$[1, 5] \times 10^{-4}$	$\{128, 256, 384\}$	2–4	$K_g \in \{4, 6, 8\}$	4.9–10.2M
ARGUS-Text^{T}	$[1, 5] \times 10^{-4}$	$\{128, 256\}$	2–3	$K_g \in \{4, 6, 8\}$	3.9–6.1M
AttriGuard-Text^{T}	—	—	—	replay budget $\{2, 3, 5\}$	0.36M ranker
TrustTrace-Terminal	$[1, 5] \times 10^{-4}$	$\{128, 256, 384\}$	2–4	$K_g \in \{4, 6, 8\}$	5.6–11.1M
TrustTrace	$[1, 5] \times 10^{-4}$	$\{128, 256, 384\}$	2–4	$K_g \in \{4, 6, 8\}$	6.2–12.4M

movement in source accuracy and ECE occurs before the default (2, 3) budget, after which the Pareto curve flattens. At the same time, the distributions for correct and incorrect candidates become more separated by three replays, with little additional separation at five or seven. The default verifier is therefore not chosen only from average latency in

Table 24; it is also the smallest budget that yields stable path ordering across stochastic replays.

APPENDIX C
COALITION-LEVEL MULTI-SOURCE VERIFICATION

For up to three suspicious sources, we evaluate a budgeted coalition verifier that replays the top candidate pairs after single-source interventions are inconclusive. Table 25 separates redundant attacks, where either source is sufficient, from complementary attacks, where both sources are required.

Pairwise replay improves coalition recall for complementary attacks but increases generator calls quadratically in the verification budget. We therefore keep coalition analysis as an audit mode and do not include it in routine graph-only screening. Mutually reinforcing attacks remain a boundary because multiple weak paths can jointly change a conclusion without any one intervention producing a large necessity score.

APPENDIX D
CALIBRATION UNDER DATASET AND GENERATION-FAMILY SHIFT

The independent Qwen3 verifier remains close to nominal coverage on in-domain workflows, but the Qwen3-generated and human-edited settings fall farther below 90%. Candidate sets broaden under shift, and selective error increases. This behavior is expected because the calibration and test workflows are no longer exchangeable. The table prevents the in-domain conformal guarantee from being generalized to new model families or human revisions and indicates that deployment requires labeled recalibration data, a conservative coverage target, or an explicit shift monitor.

APPENDIX E
BASELINE SEARCH SPACES AND PARAMETER COUNTS

The selected TrustTrace configuration uses learning rate 2×10^{-4}, width 256, depth three, dropout 0.2, and $K_g = 6$, with 7.8M trainable parameters. Every trainable comparator receives 30 validation trials and the same early-stopping rule. AttriGuard-Text does not train a graph encoder; its search varies the replay budget and the common 0.36M source ranker. Trial-level configurations, validation metrics, selected checkpoints, and parameter-count scripts were retained for reproducibility checks.

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DATA AND CODE AVAILABILITY

The public datasets used in this study, including BIPIA, InjecAgent, and AgentDojo, were obtained through their official project repositories or publication channels and processed according to their licenses and terms of use. BIPIA-Trace, InjecAgent-Text, and AgentDojo-Text are workflow-level adaptations created in this study and are not official extensions released by the original authors.

The data, code, annotations, prompts, split identifiers, model-request records, and evaluation materials supporting the findings of this study will be made available after acceptance upon reasonable request to the corresponding author, subject to applicable third-party licenses, privacy safeguards, and redistribution restrictions. The materials covered by this policy include the redistributable portions of EduTrace-IPI, CourseRAG-Audit, Qwen3-Gen-Audit, and HumanEdit-Audit; immutable train, validation, calibration, and test identifiers; source-family, duplicate-cluster, topic, attack-template, and generation-seed groupings; per-seed workflow predictions; top-k path outputs; annotator-specific and adjudicated source, stage, span, edge, ordinal-contamination, and multi-path labels; disagreement flags; GPT-5.5 and Qwen3 request records and hashes; neutralization and judging prompts; baseline search histories; bootstrap indices; Holm-family definitions; graph, training, replay, calibration, path-matching, and metric code; environment specifications; `SHA256SUMS`; and `table_provenance.json`.

Third-party or restricted source text will not be redistributed; documented official origins, stable identifiers, and reconstruction instructions will be provided where permitted. Archived model requests and outputs support record-level auditing and table regeneration, but exact service-level regeneration cannot be guaranteed after a hosted provider updates a model alias or serving stack. This availability statement therefore distinguishes record-based reproducibility from future reproducibility of the hosted service.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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